Nonlinear Finite-Element Solves and Optimization with JAX and PETSc: Exact and Approximate Second-Order Routes in a Maintained Sparse Solver Workflow

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Abstract

We present `fenics_nonlinear_energies` as a software-and-methods workflow for nonlinear finite-element solves and optimization that spans pure JAX references, FEniCS references where they exist, and a maintained distributed JAX+PETSc implementation. Across the maintained benchmark suite we cover nonlinear scalar problems (p-Laplace and Ginzburg–Landau), large-deformation hyperelasticity, two-dimensional and three-dimensional Mohr–Coulomb plasticity, and topology optimization. The methodological focus is a solver-centric combination of JAX for local differentiation with PETSc for sparse assembly, globalization, Krylov solves, and preconditioning, together with three second-order routes: element automatic differentiation, constitutive automatic differentiation, and colored sparse finite-difference recovery.

The evidence is separated into validation and performance layers. For the three-dimensional heterogeneous Plasticity3D benchmark, a same-case source-faithfulness comparison against the source-style endpoint surrogate is tight, with relative differences 3.877×10^{-5} in work, 3.517×10^{-3} in displacement, and 8.720×10^{-3} in deviatoric-strain norm. A separate fixed-lambda source-operator diagnostic uses the matched glued-bottom boundary contract and passes the configured gated quantities, with $u_{\max}(\lambda)$ relative L^2 difference 6.396×10^{-6} and endpoint displacement relative L^2 difference 8.299×10^{-4}; endpoint deviatoric-strain and boundary-profile agreement are reported as diagnostics. This validates the implemented endpoint surrogate within the tested fixed-load contract, but not as a path-consistent replacement for an incremental elastoplastic history solve. On the locked $P_4(L_1)$, $\lambda_{\mathrm{sr}} = 1.5$, 8-rank derivative ablation, constitutive AD has the lowest median wall time among the three tested routes while reaching the same terminal state as element AD and colored SFD. A serial hyperelastic companion comparison against JAX-FEM on the same mesh and displacement schedule passes a 5% fairness gate with repository-energy and field differences below 10^{-3}; this evidence is hyperelastic-only.

The largest maintained distributed timing study remains the promoted Plasticity3D study on $P_4(L_2)$ with constitutive-AD local assembly with same-mesh PMG, which attains $18.05\times$ end-to-end speedup on 32 MPI ranks relative to one rank at the same promoted nonlinear target. The topology optimization study complements this with a distributed design-and-mechanics workflow that remains operational at 768×384 resolution and 32 ranks. Within that scoped evidence, the paper's contribution is a maintained implementation pattern: derivative-rich local models, sparse nonlinear solver policy, and distributed execution can be kept together in one reproducible workflow across a heterogeneous family of FEM problems.

1 Introduction

Recent open-source finite-element and PDE-optimization frameworks cover different parts of the workflow needed for nonlinear solves and design. High-level finite-element systems such as FEniCS and `DOLFINx` emphasize expressive variational modeling together with compiled kernels and parallel assembly [1, 2]. PDE-level adjoint frameworks such as `dolfin-adjoint`, `pyadjoint`, and `cashocs` automate adjoint-centered optimization workflows and PDE-level sensitivity automation [3–5]. JAX-native or JAX-connected workflows, including JAX-FEM, second-order implicit-differentiation benchmarks for finite-element differentiable physics, AutoPDEx, JAX-CPFEM, the Firedrake–JAX bridge, and recent `FEniCSx` external-operator work, push automatic differentiation deeper into constitutive modeling and differentiable simulation [6–11]. The recently submitted JetSCI preprint is especially close in software architecture because it also couples JAX-local finite-element kernels to PETSc for distributed sparse solves [12]. Recent topology-optimization implementations such as `FEniTop` show that compact parallel design-mechanics workflows can also be built on modern `FEniCSx` infrastructure [13].

This paper targets a narrower design point than the broad differentiable-PDE landscape above. The maintained production path in `fenics_nonlinear_energies` is a solver-centric JAX+PETSc stack for distributed nonlinear finite-element solves and optimization. Local energies and constitutive laws are written in JAX, while sparse assembly, globalization, Krylov solves, and preconditioning are handled by PETSc. The goal is not to claim one universally

Family	Modeling	Differentiation	Second-order route	Parallel	Closest overlap
FEniCS DOLFINx [1, 2]	High-level variational forms	Problem-specific; AD is not the central claim	Manual or application-specific	Distributed FEM assembly and solve	Elliptic and finite-strain mechanics
dolfin-adjoint pyadjoint cashocs [3–5]	High-level PDE plus optimization loop	Adjoint-based first-order sensitivities	Not the advertised main path	MPI via the host FEM stack	PDE control, shape, and topology
JAX-FEM Xue 2026 AutoPDEx [6–8]	JAX-native PDE / FE programs	Program-level forward and reverse AD	Higher-order JAX and implicit Hessian-vector derivatives	JAX execution; PETSc optional in AutoPDEx	Nonlinear mechanics, inverse design, and FE differentiable physics
JetSCI [12]	JAX local discretizations plus PETSc sparse solves	JAX-differentiated discretization kernels	Differentiable simulation kernels with PETSc solves	JAX/GPU within node; PETSc MPI across nodes	Heterogeneous micromechanics
Firedrake–JAX FEniCSx ext. ops [10, 11]	Host FEM stack plus AD bridge	Tangent, adjoint, or local constitutive AD	Local-operator derivatives, not full sparse Hessians	Host framework parallel back end	Parameterized PDEs and constitutive models
JAX-CPFEM [9]	JAX-native crystal-plasticity FEM	Differentiable constitutive simulator	AD constitutive derivatives	GPU-oriented execution	Crystal plasticity
FEniTop [13]	FEniCSx topology code	Sensitivity-based design updates	Not a second-order solver paper	Parallel FEniCSx workflow	2D and 3D topology optimization
This work	JAX local energies plus PETSc sparse solvers	Element AD, constitutive AD, and colored SFD	Local Hessians/tangents or sparse recovery	PETSc MPI vectors, matrices, SNES, and multigrid	p-Laplace, GL, hyperelasticity, plasticity, topology

Table 1: State-of-the-art positioning of the framework families most relevant to this paper. The columns record source-supported workflow properties rather than subjective scores.

preferred framework, but to document one specific combination that is practically useful for our benchmark families: high-level local differentiation coupled to distributed sparse Newton–Krylov execution on hard nonlinear mechanics and design problems. Relative to closely related hybrid JAX–PETSc work, the emphasis here is on a maintained nonlinear energy-minimization benchmark suite and on comparing derivative routes within the same sparse solver workflow rather than on establishing a general-purpose differentiable-simulation framework.

The central claim is therefore deliberately limited. We do not claim that every benchmark requires exact Hessians, nor that mainstream frameworks cannot provide them. We instead show that several derivative routes can be made operational inside one maintained distributed workflow:

1. exact element-level derivatives when the local state dimension is still affordable;
2. constitutive-level automatic differentiation for high-order plasticity, where local tangents are much cheaper than full element Hessians;
3. colored sparse finite-difference (SFD) recovery when exact second-order information is not the right cost-quality tradeoff.

These routes are evaluated on six maintained benchmark families: p-Laplace, scalar Ginzburg–Landau, hyperelasticity, Plasticity2D, Plasticity3D, and topology optimization. The paper gives all six families top-level space because the main contribution is methodological rather than a single benchmark-specific solver. At the same time, Plasticity3D and topology optimization receive the most detailed comparison because they are the closest stress tests for the maintained JAX+PETSc path; Plasticity3D is also the family in which source-faithfulness, surrogate interpretation, and timing evidence have to be separated most carefully. The geomechanics context for that family follows the finite-element strength-reduction and 3D slope-stability literature [14–18], but the repository object studied here is explicitly the implemented endpoint surrogate, not a claim of path-consistent incremental-history equivalence.

1.1 State of the Art and Positioning

For the purposes of this paper, the most relevant comparison axes are the modeling layer, the scope of differentiation, the form in which second-order information enters the workflow, the parallel execution model, and the benchmark families closest to our maintained suite. Table 1 summarizes that comparison using conservative, source-backed descriptors.

Three comparisons matter most for the narrative that follows. First, relative to high-level FEM and adjoint ecosystems [1–5], `fenics_nonlinear_energies` moves the center of gravity from PDE-level adjoint-centered automation toward distributed nonlinear solve policy and multiple second-order routes inside the same sparse solver stack. Second, relative to JAX-native mechanics and differentiable-simulation frameworks [6–9], the main distinction is that the maintained implementation here centers PETSc-owned sparse MPI data structures and preconditioned Newton solves, whereas the cited JAX-native frameworks emphasize JAX-level modeling, local differentiation, GPU execution, second-order implicit differentiation, or differentiable-physics workflows. Third, relative to recent bridge and hybrid papers that combine JAX-style differentiation with established FEM or solver environments [10–12], the paper's emphasis is on extending that bridge-style differentiation story into benchmarked PETSc sparse assembly, Krylov solves, and nonlinear globalization for benchmark families that include topology optimization and 3D Mohr–Coulomb-type mechanics.

The resulting positioning statement is intentionally modest. `fenics_nonlinear_energies` is not presented as a general differentiable-PDE platform, nor as a replacement for the high-level FEM and adjoint ecosystems cited above. It is presented as a maintained JAX+PETSc workflow for distributed nonlinear energy minimization and equilibrium solves in which derivative fidelity, sparse linear algebra, and preconditioner policy are all part of the experimental design and comparison criteria rather than merely implementation detail.

The paper is organized as follows. Section 2 deepens the literature context behind the summary above. Section 3 describes the common optimization and derivative machinery, including line search, trust regions, and colored derivative recovery. Section 4 explains how these ideas are mapped onto pure JAX, FEniCS, and JAX+PETSc. Section 5 defines the benchmark suite, with each benchmark separated into literature model and repository specialization. Section 6 then reports the validation and external-comparison evidence, deliberately separating hyperelastic JAX-FEM agreement from the Plasticity3D source-family diagnostic ladder. Section 7 reports timing, scaling, derivative-route, and discretization-comparison evidence, followed by supporting solver evidence and a discussion of limitations.

2 Related Work

The literature most relevant to this manuscript falls into seven closely connected groups: high-level finite-element automation, PDE-level adjoint optimization, JAX-native differentiable FEM, JAX-to-FEM/PETSc bridge mechanisms, Mohr–Coulomb slope-stability source-lineage context, topology optimization workflows, and sparse nonlinear solver infrastructure. The purpose of this section is not to assign a single ranking to those groups, but to state precisely which part of each literature thread the present paper uses as context.

High-level finite-element automation. The original FEniCS project established the high-level finite-element programming model in which weak forms are expressed symbolically and compiled into efficient kernels [1]. The more recent `DOLFINx` preprint keeps that high-level mathematical abstraction while moving to a data-oriented design with explicit support for modern parallelism, custom kernels, and direct access to lower-level data structures [2]. This literature defines the reference standard for what an expressive open-source FEM environment should provide.

Adjoint-based PDE optimization. `dolfin-adjoint` showed that adjoints of high-level transient finite-element programs can be derived automatically from the symbolic problem description [3]. The later `pyadjoint` software paper documents the maintained FEniCS/Firedrake adjoint framework used to automate such workflows in practice [4]. The present `cashocs` software paper extends that PDE-constrained-optimization thread to shape optimization, optimal control, topology optimization via level sets, and MPI-enabled workflows [5]. These works are the right comparison point for adjoint-centered PDE-level automation and optimization workflows, but they do not have the same focus as the present paper on explicit second-order routes inside distributed sparse nonlinear solve stacks.

JAX-native differentiable FEM. The broad AD background is surveyed by Baydin et al. [19], while JAX provides the program-transformation environment that makes higher-order derivatives, vectorization, and just-in-time compilation readily usable in scientific computing [20]. Within finite elements, JAX-FEM shows that inverse design, nonlinear constitutive modeling, and GPU-oriented differentiable workflows can be implemented inside a JAX-native 3D FE solver [6]. Xue's later finite-element differentiable-physics paper is a direct second-order comparator because it derives implicit Hessian-vector products and evaluates Newton-CG-style use of exact Hessian information on finite-element inverse-problem benchmarks [7]. AutoPDEx broadens that JAX-native PDE picture with nonlinear minimizers, implicit differentiation, and optional PETSc integration [8]. JAX-CPFEM is especially relevant to the mechanics side of the present paper because it couples differentiable FEM, nonlinear crystal plasticity, and GPU execution in a recent open-access workflow [9]. Together, these papers define the closest modern literature context for the repository's JAX-based local differentiation layer and second-order derivative comparisons.

Bridges between JAX-style AD and established FEM stacks. The Firedrake–JAX bridge of Yashchuk [10] is important because it bypasses differentiation through low-level solver iterations by using tangent-linear and adjoint equations derived from the high-level PDE representation. The recent `FEniCSx` external-operator paper by Latyshev et al. [11] is even closer to the constitutive story of this manuscript: it uses algorithmic automatic differentiation for general constitutive models inside `FEniCSx`, including a Mohr–Coulomb example. Those papers show that constitutive- or solve-level differentiation can be combined with established FEM environments without differentiating every low-level algebraic step. The present paper extends that bridge-style differentiation story into a PETSc-centered sparse assembly and nonlinear-solve setting with explicit comparison between element AD, constitutive AD, and colored SFD. The JetSCI preprint is the closest recent hybrid in architecture: it uses JAX for local finite-element discretization and PETSc for robust distributed sparse solves in heterogeneous micromechanics [12]. We therefore treat it as evidence from a submitted preprint that the JAX-local/PETSc-global split is an active design point, while keeping the present paper's claim narrower: a maintained nonlinear energy-minimization suite with derivative-route comparisons across the benchmark families below.

Mohr–Coulomb slope stability and source-lineage context. The plasticity benchmarks use Mohr–Coulomb-type slope-stability problems as reviewer-visible stress tests, so their citations have to distinguish continuum plasticity, strength-reduction practice, and repository-specific surrogates. Davis' plasticity treatment and the finite-element strength-reduction studies of Tschuchnigg et al. [14] provide the Davis-style reduction context used in the paper. For constitutive integration, the incremental reference frame is return mapping with consistent tangents [21, 22], with Mohr–Coulomb-specific nonsmooth return operators treated by Sysala et al. [15]. The recent Sysala slope-stability line also documents variational and optimization viewpoints for shear strength reduction [16, 17] and a 3D continuation/iterative-solver source family [18]. The present manuscript uses that literature to position the benchmark family, while the numerical claims remain limited to the implemented surrogate and the repository artifacts reported below.

Topology optimization workflows. The classical SIMP and educational topology-optimization references remain important because they supply the baseline compliance, filtering, and continuation ideas used throughout the field [23–26]. For the present paper, however, the more direct modern comparator is `FEniTop`, which documents a simple 2D/3D `FEniCSx` topology-optimization implementation with parallel support [13]. That paper is relevant not because it duplicates our reduced design objective, which it does not, but because it provides a recent source-backed baseline for how a compact parallel topology-optimization code can be organized in the modern FEniCSx ecosystem.

Sparse derivative recovery and scalable nonlinear infrastructure. When exact second-order information is too expensive, sparse derivative recovery via graph coloring remains the classical alternative. Curtis, Powell, and Reid gave the foundational sparse-Jacobian viewpoint [27], while Coleman and Moré extended the approach to sparse Jacobian and Hessian estimation with graph-coloring structure [28, 29]. PETSc supplies the distributed vectors, matrices, Krylov methods, nonlinear solvers, and multigrid infrastructure needed to turn those derivative objects into scalable workflows [30, 31]. In this paper, PETSc is not treated as a mere back-end library: ownership layout, Krylov policy, nonlinear globalization, and preconditioner design are all part of the scientific comparison.

Against that background, the niche of `fenics_nonlinear_energies` can be stated precisely. It sits between high-level differentiable modeling and scalable sparse nonlinear solver engineering. The contribution is not that it replaces the ecosystems above, but that it documents one specific point in the contemporary design space: maintained distributed nonlinear FEM workflows that combine JAX-based local differentiation, multiple second-order routes, and PETSc-based sparse execution on benchmark families ranging from scalar nonlinear diffusion to topology optimization and 3D Mohr–Coulomb-type mechanics.

3 Methodology

We use a common finite-element notation across all benchmark families. The continuous domain is $\Omega \subset \mathbb{R}^d$, the conforming trial space is $V_h \subset V$, the state vector is $u_h \in V_h$, and the element-restricted state on element e is u_e. When a family is posed as an energy or reduced objective minimization, we write its discrete functional as $\mathcal{J}_h(u_h)$; when a family is posed as a sequence of equilibrium solves or by a scalar equilibrium surrogate, we write the associated scalar functional as $\Pi_h(u_h)$. Hyperelasticity is the only family in which $J = \det F$ denotes the deformation Jacobian, so we avoid the bare symbol J for generic objectives elsewhere in the paper. For finite-element labels, $P_k(L_\ell)$ denotes degree-k Lagrange elements on mesh level L_ℓ; problem-specific sections define the mesh hierarchy behind those levels when it matters for interpretation.

At the implementation level, the common sparse-assembly task is therefore one of two closely related forms:

$$\mathcal{J}_h(u_h) = \sum_{e=1}^{n_{\mathrm{el}}} \Phi_e(u_e) - f_{\mathrm{ext}}^T u_h, \tag{1}$$

or

$$\Pi_h(u_h) = \sum_{e=1}^{n_{\mathrm{el}}} \Phi_e(u_e) - f_{\mathrm{ext}}^T u_h, \tag{2}$$

where Φ_e may represent a scalar diffusion energy, a phase-field potential, a finite-strain stored-energy density, a repository scalarized plasticity surrogate, or the reduced design objective used in topology optimization. The benchmark sections specialize equations (1) and (2) to each family and define any additional symbols locally.

3.1 Finite-element and derivative structure

The main design choice in `fenics_nonlinear_energies` is that problem definitions should remain energy-centered whenever possible. This gives three derivative routes:

1. **element AD:** differentiate the full scalar element energy $\Pi_e(u_e)$ with respect to the element DOF vector;
2. **constitutive AD:** differentiate a quadrature-point energy density $\psi(\varepsilon_q)$, then assemble $r_e = \sum_q w_q B_q^T \sigma_q$ and $K_e = \sum_q w_q B_q^T C_q B_q$;
3. **colored SFD:** recover sparse Hessian information through directionally probed gradients/HVPs and graph coloring.

Element AD is the most literal route and is often the cleanest mathematically. Constitutive AD preserves the same scalar-energy source of truth but changes the differentiation boundary from the full element vector to the local strain state. This is especially effective for high-order plasticity, where the full element Hessian is often much more expensive than the 6×6 constitutive tangent. Colored SFD is used when exact second-order information is too costly or unnecessary.

3.2 Newton–Krylov globalization

The nonlinear solves in the repository are based on Newton or quasi-Newton updates, but they are not all globalized in the same way. Scalar benchmarks primarily use line-search Newton. Hyperelasticity uses a trust-region method with optional post-step line search. Some maintained experiments use a hybrid trust-region/line-search algorithm in which a reduced trust-region model provides the direction, while a bounded line search chooses the accepted step. The line-search and trust-region ingredients follow standard globalization ideas [32, Chs. 2–4]; the particular hybrid coupling used in some repository experiments is an implementation-specific combination of those textbook ingredients.

Algorithm 1 Hybrid trust-region / line-search Newton

Given x_0, trust radius Δ_0, tolerances, and callbacks for energy, gradient, and Hessian solves/matvecs
for $k = 0, 1, \ldots$ **do**
    Assemble $g_k = \nabla J(x_k)$
    if convergence criteria are satisfied **then**
        terminate
    end if
    Compute a Newton-like direction from $H(x_k)p = -g_k$
    if trust region is enabled **then**
        Build a small reduced model in the span of Newton and gradient directions
        Solve the reduced trust subproblem with radius Δ_k
        Optionally compare against a pure gradient trust step
        Restrict the line-search interval so the accepted step stays in the trust ball
    else
        Use the Newton direction directly
    end if
    Perform one-dimensional search on the chosen direction
    Compute actual and predicted reduction
    if trust region is enabled **then**
        accept/reject using $\rho = \text{ared}/\text{pred}$ and update Δ_k
    else
        accept only if the merit function decreases
    end if
end for

For the maintained 3D plasticity path promoted in this paper, the chosen policy is simpler: Newton with Armijo backtracking and a PMG-preconditioned `fgmres` linear solve [32, Sec. 3.1]. This is the maintained baseline workflow used for the flagship benchmark reported later in Section 7.5.

Algorithm 2 Armijo line search used in maintained Newton solves

Given state x, direction p, initial step α_0
Evaluate $J(x)$ and $g = \nabla J(x)$
for $m = 0, 1, \ldots, m_{\max}$ **do**
    $\alpha \leftarrow \alpha_0 \beta^m$
    $x_{\text{trial}} \leftarrow x + \alpha p$
    if $J(x_{\text{trial}}) \leq J(x) + c_1 \alpha g^T p$ **then**
        accept α
        return
    end if
end for
fall back to the smallest admissible step or to gradient-safe logic

3.3 Colored sparse finite differences

Sparse derivative recovery is built around the observation that if columns that do not interfere in the sparsity pattern are perturbed together, one gradient or HVP evaluation can recover multiple columns of the Jacobian or Hessian [27, 29]. In this repository, the local SFD path uses distance-2 coloring on the Hessian sparsity graph, which is especially effective when combined with overlap-domain assembly and owned-plus-ghost PETSc layouts.

Algorithm 3 Distributed colored Hessian recovery

Build or import local sparsity information for the free-DOF graph
Compute a distance-2 coloring of the Hessian pattern
for each color group c **do**
    Build the combined probe vector v_c
    Evaluate one gradient difference or HVP for v_c
    Scatter recovered entries into owned sparse storage
end for
Assemble the global PETSc matrix from owned COO/CSR entries

3.4 Constitutive autodiff for 3D plasticity

The promoted Plasticity3D result in this paper uses constitutive AD. Rather than applying second-order AD to the full $P4$ element vector, we write one scalar Mohr–Coulomb energy density per quadrature point,

$$\Phi_q(\varepsilon_q, \mathrm{material}_q), \tag{3}$$

differentiate it with respect to the six engineering-strain components, and assemble the element tangent from the resulting stress and constitutive tangent. This keeps the scalar-energy contract intact while reducing the cost of branchwise second-order information on high-order tetrahedra.

Algorithm 4 Constitutive AD assembly for high-order plasticity

for each local element e **do**
    for each quadrature point q **do**
        Evaluate strain $\varepsilon_q = B_q u_e$
        Evaluate scalar energy density $\Phi_q(\varepsilon_q)$
        Obtain stress $\sigma_q = \partial\Phi_q/\partial\varepsilon_q$
        Obtain constitutive tangent $C_q = \partial^2\Phi_q/\partial\varepsilon_q^2$
    end for
    Assemble $r_e = \sum_q w_q B_q^T \sigma_q$
    Assemble $K_e = \sum_q w_q B_q^T C_q B_q$
end for

The key methodological point is that automatic differentiation is applied directly to the implemented scalar branch potential. The code does not switch to a hand-derived constitutive tangent. Away from branch switching sets this yields branchwise exact derivatives of the implemented surrogate; at switching sets the model remains nonsmooth, so the resulting tangents should be read as branchwise surrogate derivatives rather than globally smooth continuum derivatives.

4 Implementation

The repository intentionally keeps three implementation strata.

Pure JAX reference paths. These are the most compact differentiable implementations in the codebase. They are useful for formulation checks, serial references, and debugging AD logic. They are not intended to be the primary production route for the largest distributed benchmarks. Their main scientific role in this paper is to show that the energy formulations used in the maintained stack can also be expressed in a compact differentiable form without sparse distributed infrastructure.

FEniCS reference paths. The FEniCS implementations remain valuable on families where they exist, especially p-Laplace, Ginzburg–Landau, and hyperelasticity. They provide a trusted high-level FEM reference and help distinguish modeling issues from solver-stack issues. In this repository they are used mainly as comparison baselines rather than as the mainline production path.

Maintained JAX+PETSc path. This is the central implementation of the paper. It combines JAX-based local physics with PETSc distributed vectors, matrices, Krylov solvers, and preconditioners. Problem-specific kernels stay in JAX; sparse distributed assembly, nonlinear globalization, and linear algebra stay in PETSc.

4.1 Autodiff modes

The maintained stack supports several second-order modes. The precise available choices depend on the benchmark family, but the common logic is:

- **element AD**: exact local Hessians from differentiating the full element energy;
- **constitutive AD**: exact local constitutive tangents assembled from quadrature-point energy densities;
- **local SFD / colored SFD**: sparse approximate Hessian recovery when exact second-order information is not the appropriate cost tradeoff.

This split is scientifically important. It lets us compare exact second-order information against carefully structured sparse approximations without changing the rest of the nonlinear solver stack. In other words, the paper is not only a comparison of frameworks, but also a comparison of derivative strategies inside one common solver infrastructure.

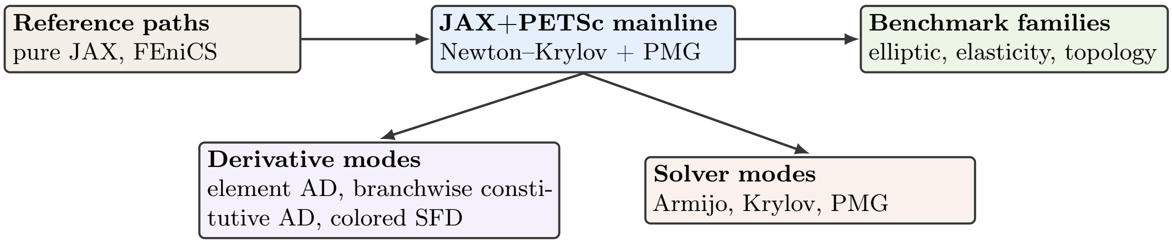


Figure 1: Software structure of the manuscript. The maintained JAX+PETSc implementation is the main production path; pure JAX and FEniCS remain reference paths where they already exist in the repository.

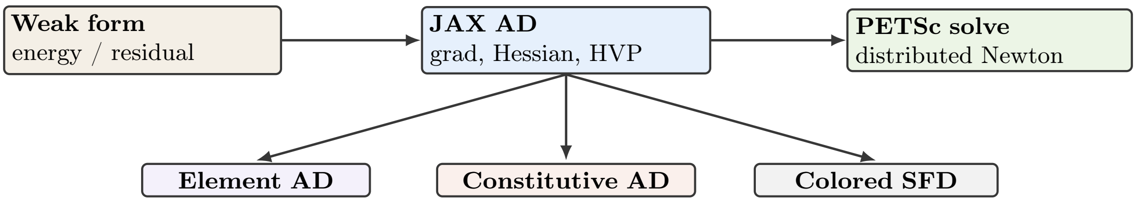


Figure 2: Derivative pathways from scalar energies and quadrature-point potentials to distributed PETSc nonlinear solves.

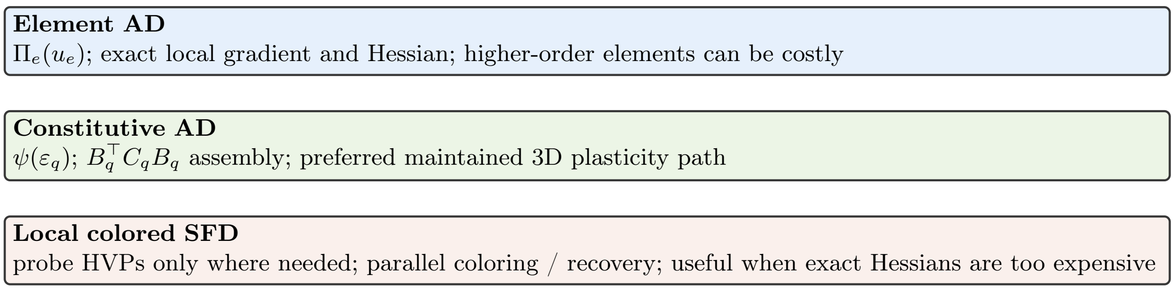


Figure 3: Second-order derivative modes exposed in the maintained stack. The important engineering choice is that the source of truth remains an energy or energy-density expression even when the tangent path changes from element AD to constitutive AD or colored SFD.

4.2 Linear solvers and preconditioners

The maintained families use different Krylov and preconditioner combinations because the linear systems are qualitatively different:

- **p-Laplace:** mostly `cg + hypre`, with exact element Hessians or local-SFD Hessians in the JAX+PETSc path.
- **Ginzburg–Landau:** `gmres + hypre`, reflecting the more indefinite linearization.
- **Hyperelasticity:** trust-region outer iterations with `gamg`-preconditioned Krylov solves.
- **Plasticity (2D):** deep-tail multigrid hierarchies whose bottlenecks are dominated by the top smoother and repeated Krylov work.
- **Plasticity3D:** local PMG on same-mesh `P4 -> P2 -> P1`, coarse `cg + hypre`, and an elastic initial guess on the same hierarchy.
- **Topology:** `fgmres + gamg` for the mechanics step, coupled to a separate distributed design update.

The comparison material later in the manuscript also includes a source-repo PMG-shell variant and alternative Krylov/preconditioner diagnostics. Those are not the primary maintained defaults, but they are informative because they reveal how much of the final performance difference comes from derivative paths, preconditioners, or outer solver logic.

4.3 Distributed assembly and coloring

The JAX+PETSc implementation uses PETSc-owned ranges, local-plus-ghost layouts, and owned-row sparse assembly. This matters for both exact and approximate second-order information. Exact element or constitutive assembly can write directly into local sparse buffers. Colored SFD requires an additional graph-coloring step, but once the coloring is known the probe evaluations can still be carried out in a rank-local manner with only scalar reductions or owned sparse insertion in the hot loop.

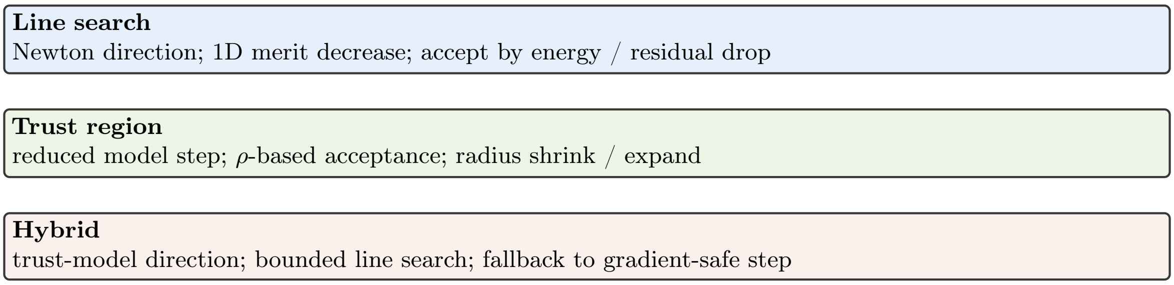


Figure 4: Globalization choices used across the repository, from pure line search to trust-region and hybrid strategies.

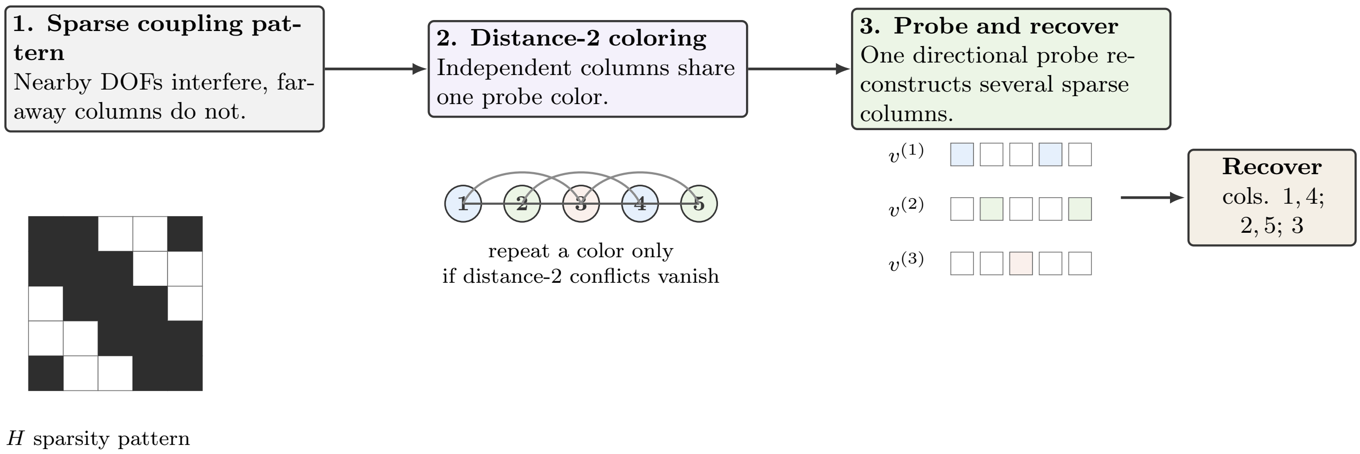


Figure 5: Schematic view of the colored sparse-recovery logic used by the local SFD Hessian path: a sparsity pattern, a distance-2 coloring, and the corresponding grouped probes used for recovery.

Family	FEniCS	pure JAX	JAX+PETSc	Higher-order / advanced AD
p-Laplace	yes	yes	yes	element AD and local colored SFD
Ginzburg–Landau	yes	no	yes	element AD and local colored SFD
Hyperelasticity	yes	yes	yes	trust-region element AD
Plasticity2D	no	no	yes	repository scalarized potential and same-mesh PMG
Plasticity3D	no	no	yes	constitutive AD, element AD, and same-mesh PMG
Topology optimization	no	yes	yes	distributed design updates and PETSc mechanics

Table 2: Implementation capability matrix across the maintained benchmark families. "Higher-order / advanced AD" summarizes the derivative choices that are scientifically interesting in each family.

Table 2 summarizes the implemented surface. The most important structural point is that the maintained production story is always the JAX+PETSc column, even when a family also has FEniCS or pure JAX references.

5 Benchmark Problems

The benchmark suite is intentionally heterogeneous. The central claim of the paper is not that one solver policy dominates every nonlinear PDE family, but that one maintained differentiable sparse-solver stack can cover scalar elliptic problems, phase-field-type energies, finite-strain mechanics, elastoplastic slope stability, and reduced topology optimization without switching conceptual frameworks.

Family	Grid / mesh	Stop rule	Compared paths	Main difficulty
p-Laplace	L_9	Newton + line search	FEniCS, pure JAX, JAX+PETSc	nonlinear elliptic solve with exact sparse Hessians
Ginzburg–Landau	L_9	Newton + line search	FEniCS, JAX+PETSc	indefinite local curvature from the double well
Hyperelasticity	L_4, 24 steps	trust-region path	FEniCS, pure JAX, JAX+PETSc	nonconvex large-deformation mechanics
Plasticity2D	$P_4(L_5)$–$P_4(L_7)$	endpoint or fixed work	JAX+PETSc only	same-mesh PMG and nonlinear tail behavior
Plasticity3D	$P_1/P_2/P_4$	$\|g\| < 0.01$	constitutive-AD and source PMG variants	heterogeneous 3D Mohr–Coulomb with constitutive AD
Topology	768×384	stall-stop continuation	pure JAX, JAX+PETSc	distributed design-mechanics coupling

Table 3: Benchmark families, stopping rules, and principal comparison paths. The later numerical section revisits the same families from the timing and scaling viewpoint.

Family	FEniCS	pure JAX	Notes
p-Laplace	yes	yes	All three stacks exist on maintained showcase cases.
Ginzburg–Landau	yes	no	FEniCS and JAX+PETSc form the maintained comparison.
Hyperelasticity	yes	yes	pure JAX is a serial formulation reference only.
Plasticity2D	no	no	The maintained story is JAX+PETSc only.
Plasticity3D	no	no	Source assembly exists as supporting comparison, not as a maintained reference path.
Topology	no	yes	Parallel fine-grid path is JAX+PETSc; pure JAX remains the serial design reference.

Table 4: Availability of FEniCS and pure JAX references. Reference paths are used where the repository already provides documented implementations; absent cells are left absent rather than filled with speculative baselines.

5.1 p-Laplace

The literature model is the standard stationary p-Laplace Dirichlet problem and its associated energy formulation [33, Ch. 2]. The unknown is a scalar field $u : \Omega \to \mathbb{R}$ and the continuous energy is

$$\mathcal{E}(u) = \int_\Omega \frac{1}{p}|\nabla u|^p \, dx - \int_\Omega f u \, dx, \tag{4}$$

whose Euler–Lagrange equation is

$$-\nabla \cdot \left(|\nabla u|^{p-2}\nabla u\right) = f \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega. \tag{5}$$

The weak problem is: find $u \in W_0^{1,p}(\Omega)$ such that

$$\int_\Omega |\nabla u|^{p-2}\nabla u \cdot \nabla v \, dx = \int_\Omega f v \, dx \qquad \forall v \in W_0^{1,p}(\Omega), \tag{6}$$

The repository specialization used in this paper takes $\Omega = (0,1)^2$, exponent $p = 3$, forcing $f = -10$, and a conforming P_1 triangular hierarchy. The discrete problem is

$$u_h \in V_h \subset W_0^{1,p}(\Omega) : \qquad u_h = \arg\min_{v_h \in V_h} \mathcal{E}_h(v_h), \tag{7}$$

where V_h is the chosen conforming finite-element subspace. This is the cleanest benchmark in the paper for checking exact gradients, exact Hessians, and sparse-recovery alternatives against each other without constitutive or geometric complications.

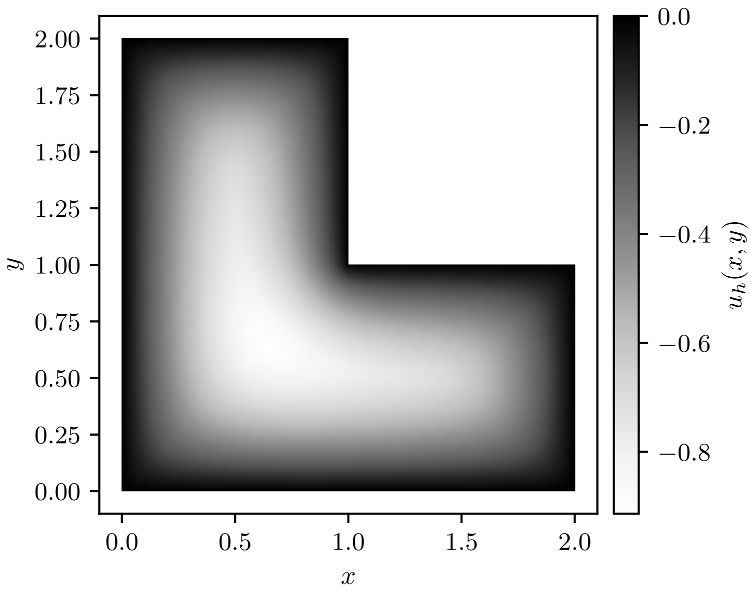


Figure 6: Representative p-Laplace state on the maintained hierarchy.

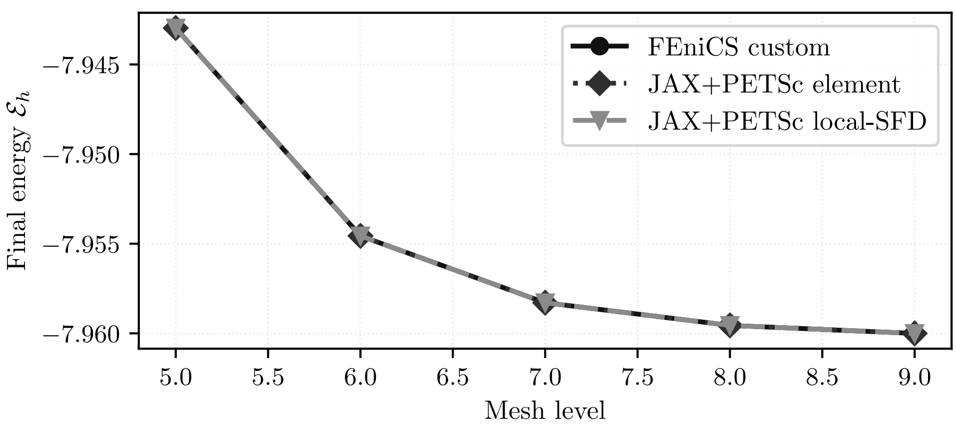


Figure 7: Final discrete energies from equation (7) across the maintained mesh hierarchy. The element-AD and local-SFD JAX+PETSc paths are essentially energy-consistent with the FEniCS reference.

Path	Energy	Newton iters	Krylov iters	Wall time [s]
FEniCS custom Newton	−7.942,968,719	5	15	0.0782
JAX+PETSc element AD	−7.942,968,719	6	17	0.0798
JAX+PETSc colored SFD	−7.942,968,711	6	6	0.194

Table 5: Representative final discrete energies from equation (7) on the showcase case.

The within-benchmark picture is therefore one of formulation agreement: figure 7 and table 5 show that the maintained routes converge to the same discrete energy from equation (7). Scaling and timing for this family are reported in Section 7.1.

5.2 Ginzburg–Landau

We cite Ginzburg and Landau [34] for the historical origin of Ginzburg–Landau free-energy modeling only. The maintained benchmark used here is a repository specialization: a scalar real-valued double-well problem on $\Omega = [-1, 1]^2$ with homogeneous Dirichlet boundary conditions and energy

$$\mathcal{E}(u) = \int_\Omega \frac{\varepsilon}{2}|\nabla u|^2 + \frac{1}{4}(u^2 - 1)^2 \, dx, \qquad \varepsilon = 10^{-2}. \tag{8}$$

Its strong form is

$$-\varepsilon\Delta u + u(u^2-1) = 0 \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega, \tag{9}$$

and the weak problem is: find $u \in H_0^1(\Omega)$ such that

$$\int_\Omega \varepsilon \nabla u \cdot \nabla v + (u^2-1)uv \, dx = 0 \qquad \forall v \in H_0^1(\Omega). \tag{10}$$

The maintained discrete problem is a conforming P_1 stationary problem on the same triangular hierarchy:

$$u_h \in V_h: \qquad D\mathcal{E}_h(u_h)[v_h] = 0 \qquad \forall v_h \in V_h. \tag{11}$$

Relative to p-Laplace, the additional difficulty here is not geometry but indefinite local curvature from the double-well potential in equation (8).

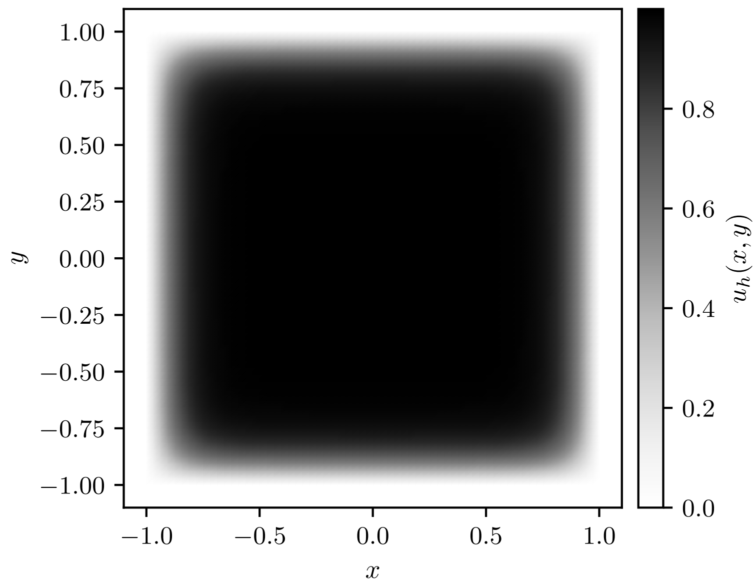


Figure 8: Representative Ginzburg–Landau state on the maintained hierarchy.

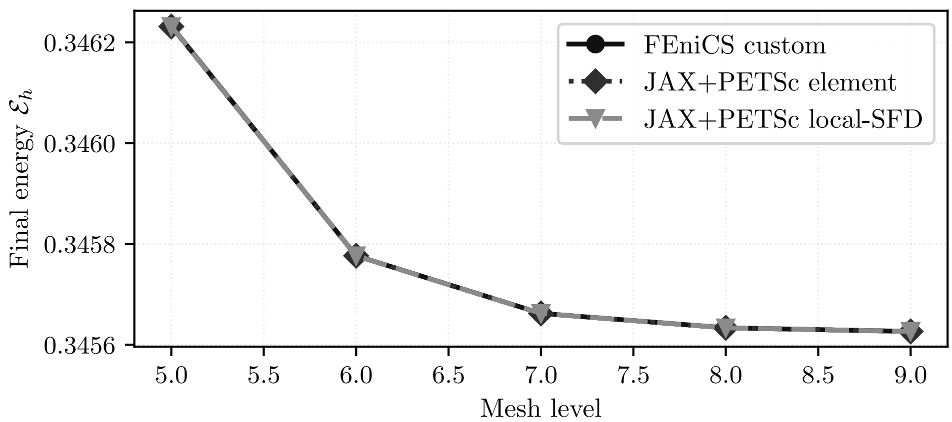


Figure 9: Final discrete energies associated with the stationary problem equation (11) across mesh levels. The maintained element-AD and local-SFD routes remain nearly coincident with the FEniCS reference.

Path	Energy	Newton iters	Krylov iters	Wall time [s]
FEniCS custom Newton	0.346,232,365,1	8	33	0.113
JAX+PETSc element AD	0.346,231,443,8	8	33	0.0949
JAX+PETSc colored SFD	0.346,231,443,8	8	33	0.247

Table 6: Representative final discrete energies associated with equation (11) on the showcase case.

The local picture is again one of formulation consistency, now for a more solver-sensitive nonconvex energy. Strong scaling and solver-time comparisons for this family are reported in Section 7.2.

5.3 Hyperelasticity

The literature model is a standard finite-strain hyperelastic equilibrium problem written in terms of deformation map $y(X) = X + u(X)$, deformation gradient $F = \nabla y$, Jacobian $J = \det F$, and first Piola stress $P = \partial W/\partial F$ [35, Chs. 4–6]. The repository specialization used in this paper is a cantilever benchmark with the compressible Neo-Hookean-type density

$$\Pi(y;\eta) = \int_\Omega C_1 \left(\operatorname{tr}(F^T F) - 3 - 2\ln J\right) + D_1(J-1)^2\, dx, \tag{12}$$

on $\Omega = [0, 0.4] \times [-0.005, 0.005]^2$ with $C_1 \approx 3.846 \times 10^7$ and $D_1 \approx 8.333 \times 10^7$. The left face is clamped, while the right face carries the prescribed rotating Dirichlet load path used throughout the repository, parametrized by load step η. The strong form is

$$-\nabla \cdot P(F) = 0 \quad \text{in } \Omega, \qquad y = \bar{y}(\eta) \quad \text{on } \Gamma_D, \qquad P(F)n = 0 \quad \text{on } \Gamma_N, \tag{13}$$

where $P = \partial W/\partial F$ is the first Piola stress. The weak form is: find $u \in V_{\bar{y}(\eta)}$ such that

$$\int_\Omega P(F(u)) : \nabla v\, dx = 0 \qquad \forall v \in V_0. \tag{14}$$

The discrete problem solved in the paper is the sequence of stationary points

$$u_h^{(k)} \in V_h : \qquad D\Pi_h(u_h^{(k)}; \eta_k)[v_h] = 0 \qquad \forall v_h \in V_{h,0}, \qquad k = 1, \ldots, 24, \tag{15}$$

with first-order tetrahedral vector elements and the benchmark-specific boundary data described above. This is the first family in which globalization is more scientifically important than derivative parity alone.

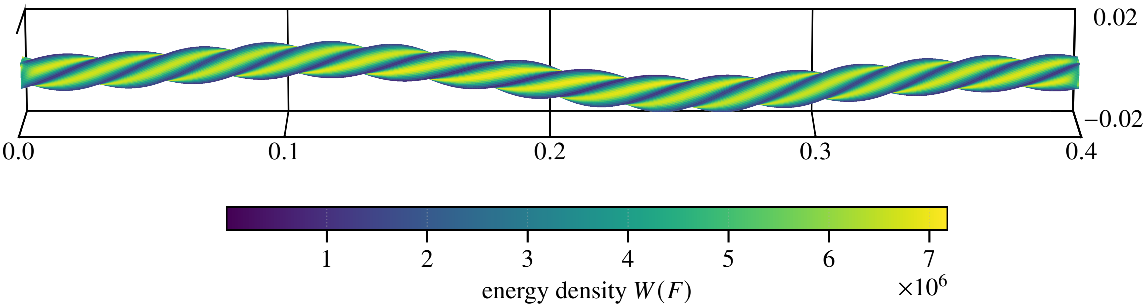


Figure 10: Representative finite-strain hyperelastic state, colored by the energy density in equation (12).

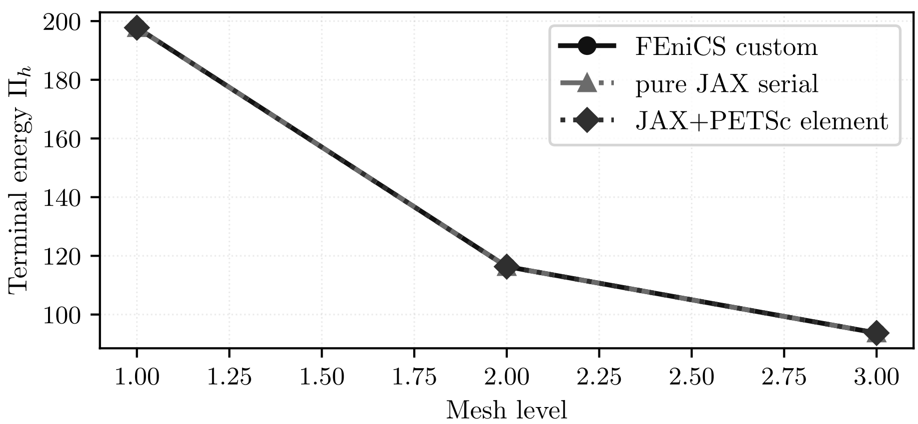


Figure 11: Final discrete energies along the stationary sequence equation (15) across the maintained mesh hierarchy. The publication story is less about exact equality than about tracking the same 24-step trajectory consistently.

Path	Energy	Steps	Krylov iters	Wall time [s]
FEniCS custom Newton	197.775,074	24	8722	6.51
JAX+PETSc element AD	197.754,582	24	10595	9.36
serial JAX	197.749,509	24	2284	41.5

Table 7: Representative 24-step terminal energies along equation (15).

The within-benchmark takeaway is therefore a globalization result: figures 10 and 11 show the terminal deformed state and terminal energies, while the scaling evidence is deferred to Section 7.3. The separate validation section adds one serial companion comparison against JAX-FEM on the same mesh and prescribed displacement schedule, with repository-energy post-comparison; it is included as a fairness-first hyperelastic implementation comparison rather than as a broad framework ranking exercise.

5.4 Plasticity2D

The literature model is a quasi-static plane-strain Mohr–Coulomb slope stability problem posed as equilibrium together with a constitutive update and history variables [14–16, 21, 22, 36]. The unknown is the displacement field $u = (u_x, u_y) : \Omega \to \mathbb{R}^2$ on the maintained homogeneous slope geometry. At increment $n+1$, the continuum equilibrium equations are

$$-\nabla \cdot \sigma^{n+1} = b^{n+1} \quad \text{in } \Omega, \qquad u^{n+1} = \bar{u}^{n+1} \quad \text{on } \Gamma_D, \qquad \sigma^{n+1} n = t^{n+1} \quad \text{on } \Gamma_N, \tag{16}$$

where σ^{n+1} and the internal variables are obtained from the elastoplastic update driven by $\varepsilon(u^{n+1})$ and the previous state. The corresponding weak form is

$$\int_\Omega \sigma^{n+1} : \varepsilon(v)\, dx = \int_\Omega b^{n+1} \cdot v\, dx + \int_{\Gamma_N} t^{n+1} \cdot v\, ds \qquad \forall v \in V_0. \tag{17}$$

The maintained runs reported here are endpoint solves, not path-consistent incremental histories. Accordingly, the repository benchmark uses the following scalarized discrete surrogate functional for differentiation and comparison:

$$\Pi_h^{2\mathrm{D}}(u_h; \lambda_{\mathrm{sr}}) = \sum_{e=1}^{n_{\mathrm{el}}} \sum_{q=1}^{n_q} w_{eq}\, \Phi_{\mathrm{MC,2D}}\left(B_{eq} u_e, \varepsilon_{eq}^{p,\mathrm{old}}, E, \nu, c_{\lambda,eq}, \phi_{\lambda,eq}, \psi_{\lambda,eq}\right) - f_{\mathrm{ext}}^T u_h, \tag{18}$$

where λ_{sr} denotes the strength-reduction factor, distinct from the Lamé parameter used later in three dimensions.

The maintained code keeps a Davis-B-style reduction path to map the raw cohesion and angles to reduced values under strength reduction; with c_0, ϕ, ψ denoting the unreduced cohesion, friction angle, and dilation angle, respectively, the benchmark uses

$$\begin{aligned} c_1 &= c_0/\lambda_{\mathrm{sr}}, & \phi_1 &= \arctan(\tan\phi/\lambda_{\mathrm{sr}})\,, & \psi_1 &= \arctan(\tan\psi/\lambda_{\mathrm{sr}})\,, \\ \beta &= \frac{\cos\phi_1 \cos\psi_1}{1 - \sin\phi_1 \sin\psi_1}, & c_\lambda &= \beta c_1, & \phi_\lambda &= \arctan(\beta \tan\phi_1)\,, \end{aligned} \tag{19}$$

with ψ_λ defined analogously when the non-associated flow rule is activated [14, 16, 36]. The showcase endpoint runs in this paper use $\lambda_{\mathrm{sr}} = 1$ and $\psi = 0°$, so the reported states should be read as repository endpoint surrogates rather than path-consistent history updates.

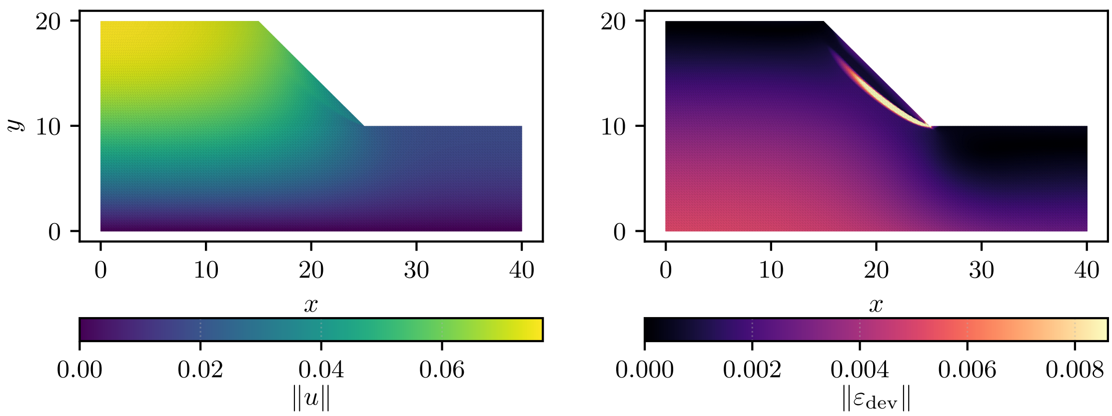


Figure 12: Representative Plasticity2D state: displacement magnitude and deviatoric strain derived from the scalar surrogate equation (18).

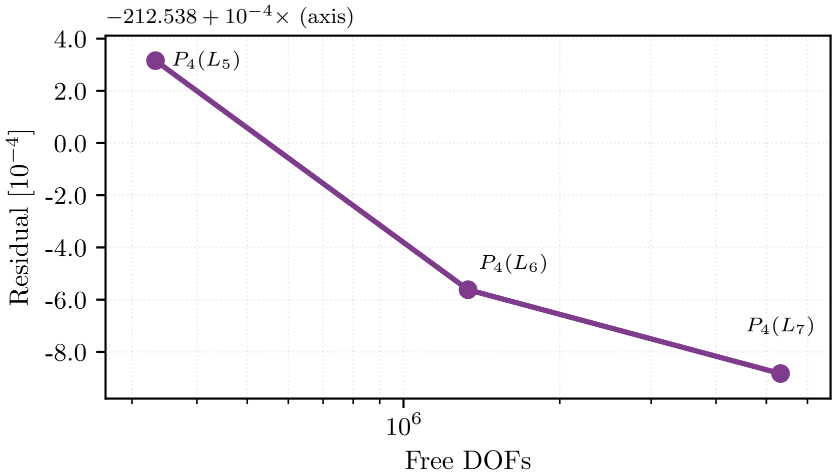


Figure 13: Resolution trend for the maintained Plasticity2D family. The plotted terminal energies are evaluations of equation (18) on the curated $P_4(L_5)$ showcase and on the larger fixed-work $P_4(L_6)$ and $P_4(L_7)$ diagnostics, which are not used as endpoint-convergence evidence.

Case	Free DOFs	Energy	Wall time [s]	Note
$P_4(L_5)$	332,960	−212.538	147	curated showcase
$P_4(L_6)$	1,331,520	−212.539	147	fixed-work at 8 ranks
$P_4(L_7)$	5,325,440	−212.539	253	fixed-work at 16 ranks

Table 8: Representative Plasticity2D endpoint values from equation (18).

This family is where multigrid hierarchy design first becomes dominant in the paper. Only the curated $P_4(L_5)$ row is treated as a completed endpoint solve; the larger $P_4(L_6)$ and $P_4(L_7)$ rows are fixed-work diagnostics for solver behavior. Solver-engineering evidence for Plasticity2D is reported later in Sections 7.4 and 8.

5.5 Plasticity3D

The literature model is again quasi-static Mohr–Coulomb equilibrium with constitutive/history updates, now on a three-dimensional heterogeneous slope geometry [14–18, 21, 22, 36]. The unknown is a displacement field $u = (u_x, u_y, u_z) : \Omega \to \mathbb{R}^3$, gravity acts in the source y-direction. At increment $n+1$, the continuum equilibrium equations are

$$-\nabla \cdot \sigma^{n+1} = \rho g_y^{n+1} e_y \quad \text{in } \Omega, \qquad u_i^{n+1} = \bar{u}_i^{n+1} \quad \text{on } \Gamma_{D,i}, \qquad \sigma^{n+1} n = 0 \quad \text{on } \Gamma_N, \tag{20}$$

where σ^{n+1} and the internal variables are obtained from the elastoplastic update driven by $\varepsilon(u^{n+1})$ and the previous state. The corresponding weak form is

$$\int_\Omega \sigma^{n+1} : \varepsilon(v)\, dx = \int_\Omega \rho g_y^{n+1} e_y \cdot v\, dx \qquad \forall v \in V_0. \tag{21}$$

The repository specialization studied in this paper uses the imported SSR mesh hierarchy with same-mesh tetrahedral $P1/P2/P4$ spaces, source component-wise boundary labels, and an additional glued-bottom condition that fixes every node with $|y| \le 10^{-12}$ in all three displacement components. We write $P_k(L_\ell)$ for polynomial degree k on mesh level ℓ; the repository mesh identifiers `L1`, `L1_2`, `L1_2_3`, and `L1_2_3_4` are displayed in the manuscript as $L_1, \dots, L_4$. The figures below use the corrected glued-bottom $P_4(L_2)$ showcase at $\lambda_{\mathrm{sr}} = 1.55$ and the finished nine-case maintained study at the same load factor. The promoted $\lambda_{\mathrm{sr}} = 1.0$ strong-scaling study is kept later in the paper as separate historical timing evidence. For those endpoint and continuation studies, the maintained code evaluates the following algorithmic scalar surrogate:

$$\Pi_h^{3\mathrm{D}}(u_h; \lambda_{\mathrm{sr}}) = \sum_{e=1}^{n_{\mathrm{el}}} \sum_{q=1}^{n_q} w_{eq}\, \Phi_{\mathrm{MC,3D}}(\varepsilon_{eq}(u_e), \bar{c}_{\lambda,eq}, \sin\phi_{\lambda,eq}, \mu_{eq}, K_{eq}, \lambda_{\mathrm{L},eq}) - f_{\mathrm{ext}}^T u_h, \tag{22}$$

where λ_{sr} is again the strength-reduction factor and λ_{L} is the Lamé parameter of the elastic predictor.

The code uses the engineering-strain ordering

$$\varepsilon = \begin{bmatrix} \varepsilon_{xx} & \varepsilon_{yy} & \varepsilon_{zz} & \gamma_{xy} & \gamma_{yz} & \gamma_{xz} \end{bmatrix}^T, \tag{23}$$

which is first mapped to tensor shear before principal values are formed. The benchmark keeps the Davis-B reduction explicitly at quadrature points:

$$
\begin{aligned}
c_{1,q} &= c_{0,q}/\lambda_{\mathrm{sr}}, & \phi_{1,q} &= \arctan(\tan\phi_q/\lambda_{\mathrm{sr}})\,, & \psi_{1,q} &= \arctan(\tan\psi_q/\lambda_{\mathrm{sr}})\,, \\
\beta_q &= \frac{\cos\phi_{1,q}\cos\psi_{1,q}}{1-\sin\phi_{1,q}\sin\psi_{1,q}}, & c_{\lambda,q} &= \beta_q c_{1,q}, & \phi_{\lambda,q} &= \arctan(\beta_q\tan\phi_{1,q})\,,
\end{aligned}
\tag{24}
$$

with

$$
\bar{c}_{\lambda,q} = 2c_{\lambda,q}\cos\phi_{\lambda,q}, \qquad \sin\phi_{\lambda,q} = \sin(\phi_{\lambda,q}), \tag{25}
$$

which are the scalar constitutive parameters actually stored in the maintained 3D assembly path [14, 16, 36]. Here ϕ denotes friction angle and ψ denotes dilation angle; the paper uses those symbols only in that geomechanics sense.

The local constitutive surrogate follows the piecewise branch structure used in the repository. Let $\varepsilon_1 \geq \varepsilon_2 \geq \varepsilon_3$ be the principal strains and let

$$
I_1 = \varepsilon_{xx} + \varepsilon_{yy} + \varepsilon_{zz}, \qquad q(\varepsilon) = \varepsilon_{xx}^2 + \varepsilon_{yy}^2 + \varepsilon_{zz}^2 + \frac{1}{2}\left(\gamma_{xy}^2 + \gamma_{yz}^2 + \gamma_{xz}^2\right). \tag{26}
$$

The elastic branch is

$$
\Phi_{\mathrm{el}} = \frac{1}{2}\lambda_{\mathrm{L}} I_1^2 + \mu q(\varepsilon), \tag{27}
$$

and the source-style return variables are

$$
\begin{aligned}
f_{\mathrm{tr}} &= 2\mu((1+\sin\phi_\lambda)\varepsilon_1 - (1-\sin\phi_\lambda)\varepsilon_3) + 2\lambda_{\mathrm{L}}\sin\phi_\lambda I_1 - \bar{c}_\lambda, \\
\gamma_{\mathrm{sl}} &= \frac{\varepsilon_1-\varepsilon_2}{1+\sin\phi_\lambda}, \qquad \gamma_{\mathrm{sr}} = \frac{\varepsilon_2-\varepsilon_3}{1-\sin\phi_\lambda},
\end{aligned}
\tag{28}
$$

with the remaining left, right, and apex thresholds defined as in the code through the same principal-strain ordering. The scalar potential is then the piecewise function

$$
\Phi_{\mathrm{MC,3D}} = \begin{cases}
\Phi_{\mathrm{el}}, & f_{\mathrm{tr}} \leq 0, \\
\Phi_{\mathrm{s}}, & \Lambda_{\mathrm{s}} \leq \min(\gamma_{\mathrm{sl}}, \gamma_{\mathrm{sr}}), \\
\Phi_{\mathrm{l}}, & \gamma_{\mathrm{sl}} < \gamma_{\mathrm{sr}} \text{ and } \gamma_{\mathrm{sl}} \leq \Lambda_{\mathrm{l}} \leq \gamma_{\mathrm{la}}, \\
\Phi_{\mathrm{r}}, & \gamma_{\mathrm{sr}} < \gamma_{\mathrm{sl}} \text{ and } \gamma_{\mathrm{sr}} \leq \Lambda_{\mathrm{r}} \leq \gamma_{\mathrm{ra}}, \\
\Phi_{\mathrm{a}}, & \text{otherwise},
\end{cases} \tag{29}
$$

where the branch energies Φ_{s}, Φ_{l}, Φ_{r}, and Φ_{a} are the repository's branchwise quadratic corrections to equation (27). The maintained constitutive-AD path differentiates the smooth branches of equation (29) directly with JAX rather than substituting a hand-derived return-map tangent; the branch interfaces remain nonsmooth and are treated as part of the algorithmic constitutive surrogate rather than as a globally smooth continuum potential.

For the strain visualizations in this paper we use the deviatoric norm

$$
\|\varepsilon_{\mathrm{dev}}\| = \sqrt{\varepsilon^T\left(\mathrm{diag}(1,1,1,\tfrac{1}{2},\tfrac{1}{2},\tfrac{1}{2}) - \frac{1}{3}mm^T\right)\varepsilon}, \qquad m = (1,1,1,0,0,0)^T, \tag{30}
$$

which matches the engineering-shear convention of equation (23).

Because equations (22) and (29) define an algorithmic endpoint surrogate rather than a path-consistent incremental history functional, the results section reports this family with a separate validation ladder. Same-case source-faithfulness to the source-style surrogate implementation is treated separately from a fixed-lambda source-operator comparison. The latter checks source-family behavior under matched geometry, materials, Davis-B reduction parameters, load schedule, and glued-bottom free-DOF mask, but it is not a path-consistent incremental-history validation. The paper therefore keeps the Plasticity3D claims at the level of a repository-specific endpoint surrogate rather than a path-consistent replacement. The related return-mapping, optimization, convex-optimization, and continuation literature [15–18] is used as source-family context, not as evidence that this repository surrogate is a full incremental elastoplastic solver.

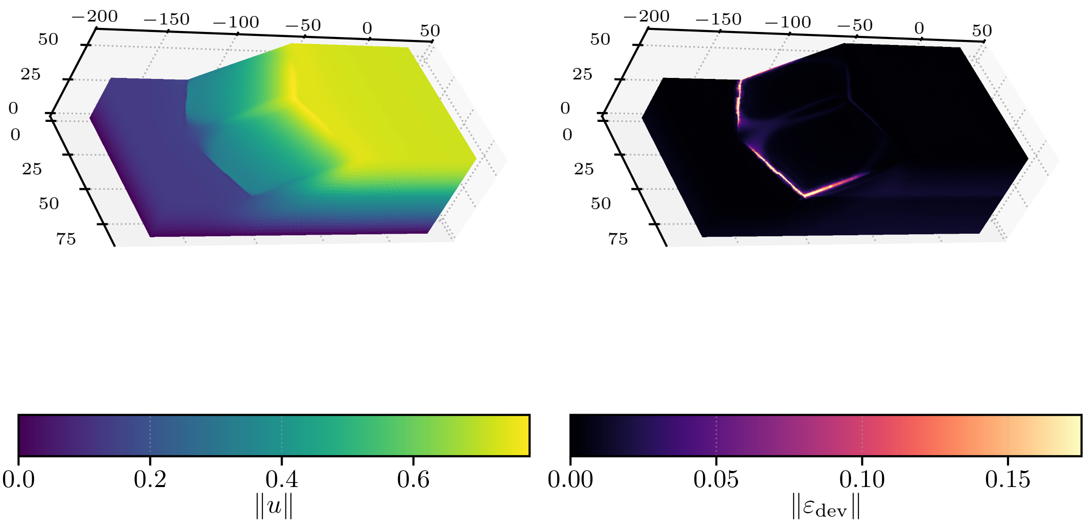


Figure 14: Corrected glued-bottom Plasticity3D showcase at $\lambda_{\mathrm{sr}} = 1.55$: displacement magnitude and surface deviatoric strain for the maintained $P_4(L_2)$ run, both derived from the scalar surrogate in equation (22).

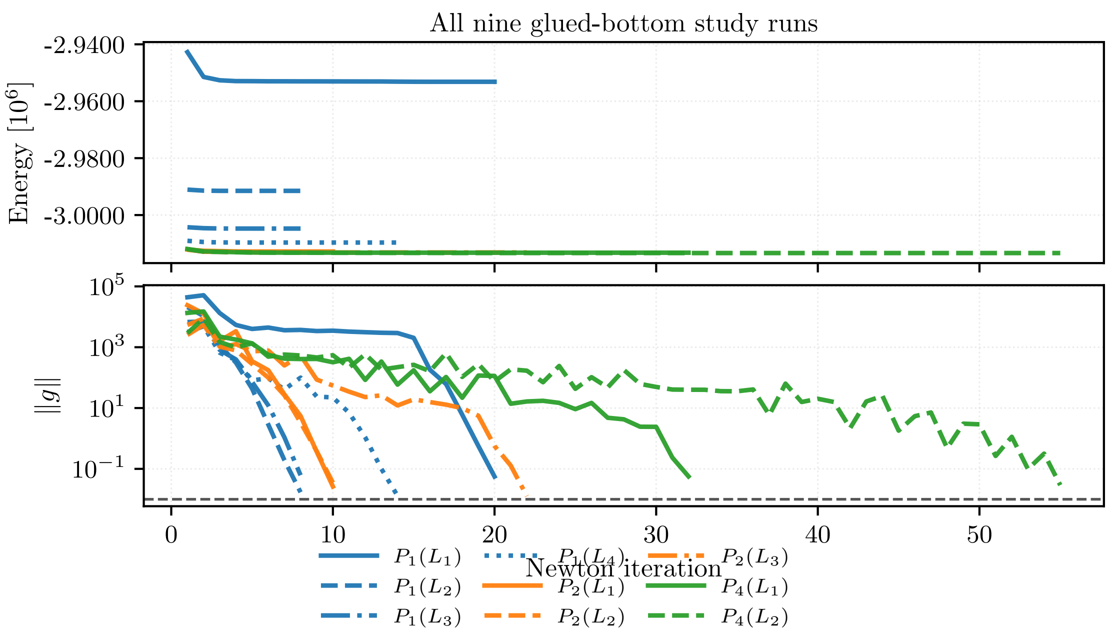


Figure 15: Newton-history convergence for all nine completed corrected glued-bottom maintained Plasticity3D study runs at $\lambda_{\mathrm{sr}} = 1.55$. The upper panel shows the value of the algorithmic surrogate from equation (22); the lower panel shows the gradient norm used as the nonlinear stopping metric, with color indicating degree and line style indicating mesh level.

Element	Free DOFs	Energy	$\|g\|_{\text{final}}$	Wall time [s]
$P_1(L_1)$	10,526	−2,953,167	4.56×10^{-3}	1.79
$P_1(L_2)$	79,024	−2,991,497	1.64×10^{-3}	8.72
$P_1(L_3)$	610,964	−3,004,742	4.07×10^{-3}	17.2
$P_1(L_4)$	4,801,816	−3,009,658	1.16×10^{-3}	96.7
$P_2(L_1)$	79,024	−3,012,813	2.47×10^{-3}	18.3
$P_2(L_2)$	610,964	−3,013,032	2.96×10^{-3}	98.3
$P_2(L_3)$	4,801,816	−3,013,149	1.14×10^{-3}	1330
$P_4(L_1)$	610,964	−3,013,227	5.53×10^{-3}	208
$P_4(L_2)$	4,801,816	−3,013,348	4.42×10^{-3}	2298

Table 9: All nine completed glued-bottom maintained Plasticity3D study rows at $\lambda_{\text{sr}} = 1.55$, reported as endpoint values of equation (22).

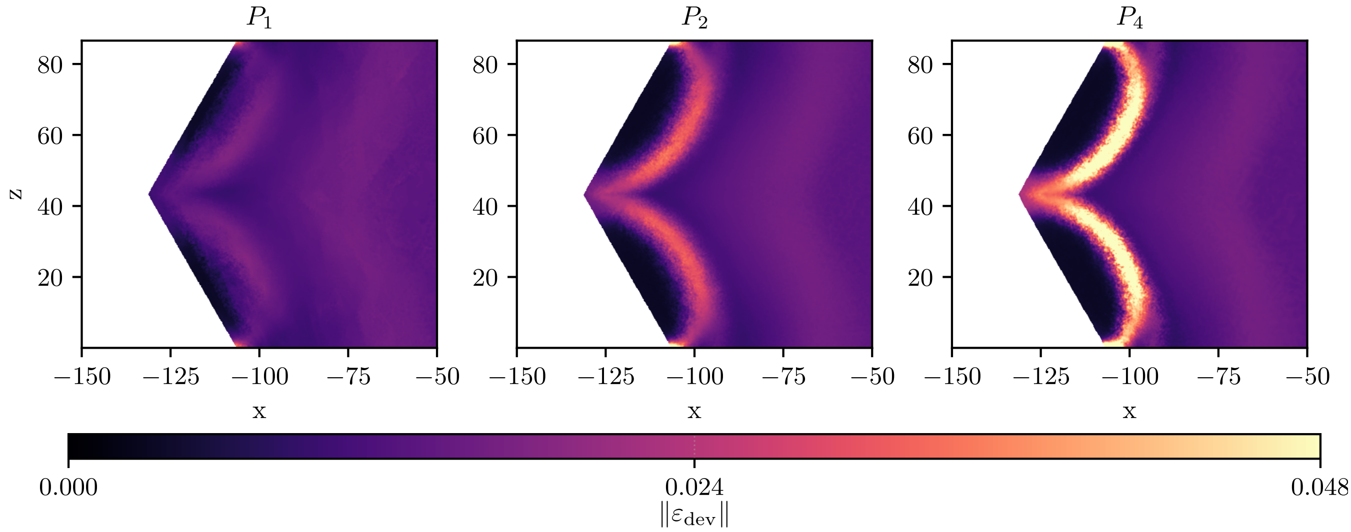


Figure 16: Shared-scale upper y-slice comparison of $\|\varepsilon_{\text{dev}}\|$ from equation (30) at the highest mesh level of each degree line: $P_1(L_4)$, $P_2(L_3)$, and $P_4(L_2)$. The three runs use the corrected glued-bottom maintained solver stack at $\lambda_{\text{sr}} = 1.55$ and are compared on one common color scale.

Plasticity3D is the benchmark where detailed constitutive definitions are most important: the paper later plots values of equation (22) and equation (30), so both quantities are defined explicitly here. Timing, scaling, and degree-versus-resolution evidence for this family are reported in Section 7.5.

5.6 Topology Optimization

The literature model is the standard SIMP-style compliance topology-optimization setting on a cantilever domain, together with filtering and continuation ideas from the classical topology literature [23–26]. In the repository specialization studied here, the topology benchmark is a reduced optimization problem rather than a single equilibrium minimization. The design variable is a latent nodal field $z_h \in Z_h$, mapped to a physical density through the logistic transform

$$\theta_h(z_h) = \theta_{\min} + (1 - \theta_{\min})\sigma(z_h), \qquad \sigma(z) = \frac{1}{1 + e^{-z}}. \tag{31}$$

For fixed density, the maintained discretization solves for the linear elastic displacement state $u_h(\theta_h)$ solving

$$a_{\theta_h}(u_h, v_h) := \sum_{e=1}^{n_{\text{el}}} |e|\, \theta_e^p \varepsilon(u_h)^T C \varepsilon(v_h) = \ell(v_h) \qquad \forall v_h \in V_{h,0}, \tag{32}$$

on the cantilever domain $\Omega = (0, 2) \times (0, 1)$ with clamped left edge and prescribed traction on a right-edge patch. The reduced objective solved in the maintained code is

$$\mathcal{J}_h(z_h) = \sum_{e=1}^{n_{\text{el}}} |e| \left[e_e^{\text{frozen}} \theta_e^{-p} + \lambda_V \theta_e + \alpha_{\text{reg}} \left(\frac{\ell_{\text{pf}}}{2} |\nabla \theta_e|^2 + \frac{1}{\ell_{\text{pf}}} W(\theta_e) \right) + \frac{\mu_{\text{move}}}{2} (z_e - z_e^{\text{old}})^2 \right], \tag{33}$$

with $W(\theta) = \theta^2(1 - \theta)^2$, SIMP-style continuation in p, and repository regularization terms that play a mesh-control role analogous to filtering/regularization in Bourdin [26]. Here e_e^{frozen} denotes the element elastic energy density from the previous mechanics state; the latent logistic map, frozen-energy term, and move penalty are repository-specific design choices rather than standard SIMP notation. The maintained outer loop alternates mechanics solves for equation (32) with reduced design updates for equation (33).

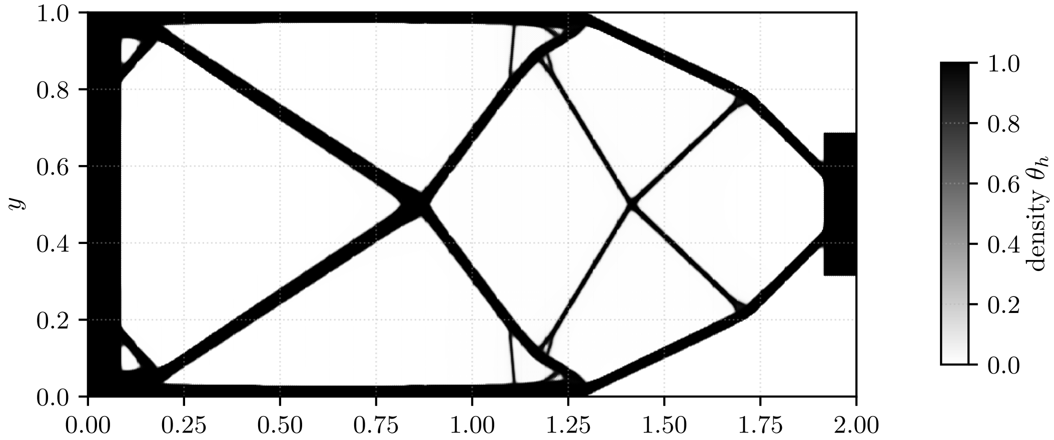


Figure 17: Representative final density field for the maintained parallel topology benchmark.

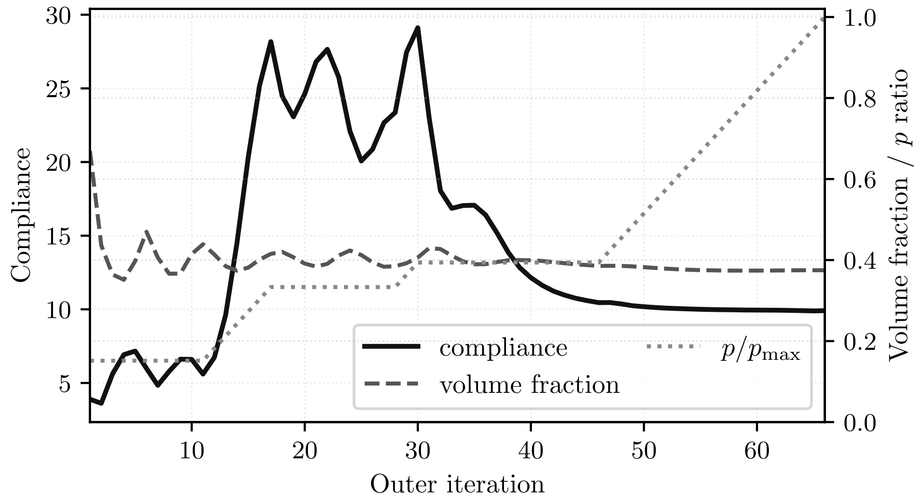


Figure 18: Objective-history view for the reduced topology objective equation (33), with compliance on the left axis and volume fraction plus the continuation schedule in p on the right axis.

Case	Ranks	Outer iters	Compliance	Volume fraction	Wall time [s]
192×96	1	121	4.1557	0.4	10
768×384	32	66	9.9074	0.3749	25.1

Table 10: Representative topology endpoint values from equation (33).

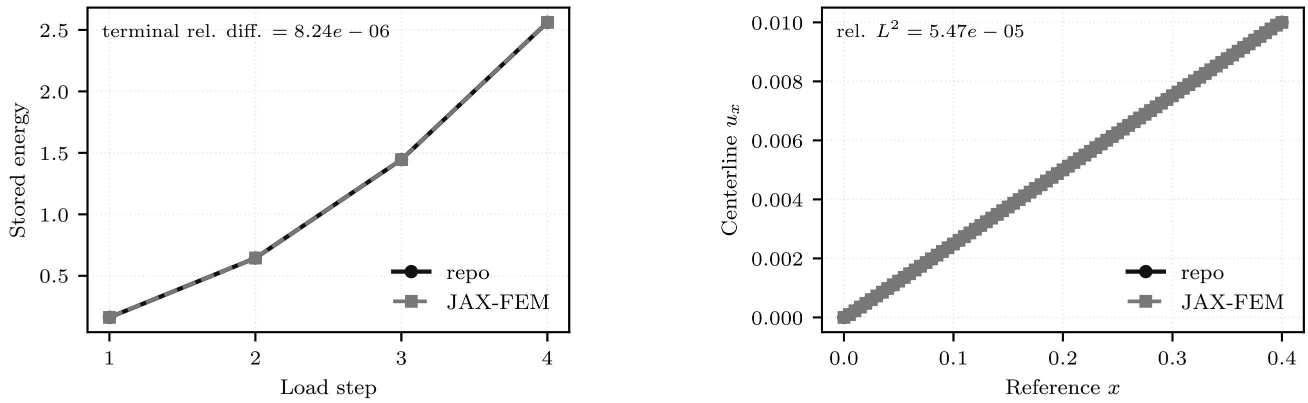
For topology, the relevant notion of "convergence" is the reduced objective history rather than a single equilibrium energy. Parallel timing and scaling for the same family are reported in Section 7.6.

6 Validation and External Comparisons

This section separates validation and external-comparison evidence from the performance results that follow. The distinction is important because the paper uses different levels of evidence for different claims: Hyperelasticity has a narrow external companion comparison against JAX-FEM, while Plasticity3D has a source-family validation ladder for the implemented endpoint surrogate.

6.1 Hyperelasticity Companion Comparison

The hyperelasticity problem definition, repository energy, and load path are defined in Section 5.3 and equation (15). The external comparison here is deliberately narrow. It uses a serial companion case against JAX-FEM on the same mesh and prescribed displacement schedule, then post-compares both terminal states with the repository energy from equation (12). This supports a solution-level implementation comparison for this hyperelastic problem only. It should not be read as a plasticity validation, a broad framework ranking, or constitutive-law identity between the two implementations: the JAX-FEM worker uses a logarithmic-J Piola form, whereas the repository energy uses the $D_1(J-1)^2$ volumetric penalty in equation (12).



(a) Terminal energy over the matched displacement schedule.

(b) Final centerline u_x profile on the common sample path.

Figure 19: Serial hyperelastic companion comparison against JAX-FEM on the same mesh and prescribed displacement schedule, with repository-energy post-comparison of the terminal states.

Group	Quantity	Relative difference	Median wall time [s]	Status
Agreement	final energy	8.24×10^{-6}	–	–
Agreement	full-field displacement relative L^2	2.01×10^{-4}	–	–
Agreement	centerline relative L^2	5.47×10^{-5}	–	–
Agreement	$u_{\max}$ curve relative L^2	8.85×10^{-16}	–	–
Timing	this work serial direct	–	0.176	–
Timing	JAX-FEM UMFPACK serial	–	2.52	–
Contract/gate	same mesh path	–	–	pass
Contract/gate	same displacement schedule	–	–	pass
Contract/gate	agreement thresholds	–	–	pass
Contract/gate	repository-energy post-comparison	–	–	used

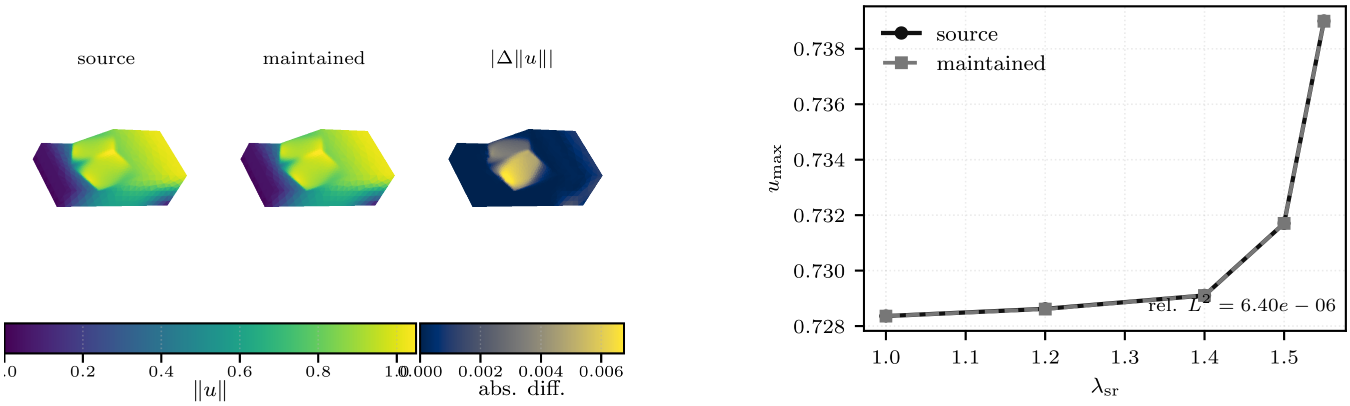
Table 11: Fairness-gated hyperelastic companion comparison against JAX-FEM.

The fairness gate in Table 11 passes on all checked quantities. The final energy relative difference is 8.244×10^{-6}, the full-field displacement relative L^2 difference is 2.011×10^{-4}, the centerline relative L^2 difference is 5.473×10^{-5}, and the $u_{\max}$-curve relative L^2 difference is effectively zero. On this small serial CPU companion case, the maintained direct solve is faster (0.176 s median wall time versus 2.524 s for JAX-FEM), but the scientific role of this table is solution agreement under a narrow hyperelastic comparison contract rather than framework ranking.

6.2 Plasticity3D Validation Ladder

The Plasticity3D benchmark is tied to a Mohr–Coulomb slope-stability source family, but the repository object studied here is the algorithmic endpoint surrogate defined in Section 5.5 and equations (22) and (29). The Sysala return-mapping, optimization, convex-optimization, and advanced continuation papers provide source-family and method context for this problem class [15–18]; they are not used here as verified numeric paper baselines.

The validation ladder therefore separates two questions. Layer 1A tests same-case source-faithfulness of the endpoint surrogate on the direct-branch $P_2(L_1)$ glued-bottom case. This layer is tight: the work relative difference is 3.877×10^{-5}, the displacement relative L^2 difference is 3.517×10^{-3}, and the deviatoric-strain relative L^2 difference is 8.720×10^{-3}. Layer 2 is different: it is a fixed-λ diagnostic against a source-operator comparison, not proof of path-consistent incremental-history equivalence. After matching the source boundary contract to the maintained glued-bottom free-DOF mask, the validation schedule is capped at the main-figure load level $\lambda_{\text{sr}} = 1.55$. It keeps the highest-successful-λ schedule proxy aligned on the tested grid and passes the configured gates for the $u_{\max}(\lambda)$ curve and endpoint displacement: the $u_{\max}(\lambda)$ relative L^2 difference is 6.396×10^{-6}, and the endpoint displacement relative L^2 difference is 8.299×10^{-4}. The endpoint deviatoric-strain relative L^2 difference (4.307×10^{-3}) and boundary-profile relative L^2 difference (5.229×10^{-4}) are reported as diagnostics rather than acceptance gates. This comparison uses a matched mesh, material table, Davis-B strength reduction, load schedule, and glued-bottom free-DOF mask, but it remains a fixed-load endpoint check rather than a path-consistent plastic-history reference.



(a) Layer 1A: same-case source-faithfulness on the surrogate.

(b) Layer 2: fixed-λ source-operator diagnostic.

Figure 20: Plasticity3D validation ladder. The left panel documents tight same-case source-faithfulness for the implemented endpoint surrogate; the right panel shows close fixed-λ source-operator agreement after matching the glued-bottom boundary contract.

Layer	Comparison	Relative difference	Status
1A	work	3.88×10^{-5}	–
1A	displacement relative L^2	3.52×10^{-3}	–
1A	deviatoric-strain relative L^2	8.72×10^{-3}	–
2	highest-successful λ	0	pass
2	$u_{\max}(\lambda)$ relative L^2	6.4×10^{-6}	pass
2	endpoint displacement relative L^2	8.3×10^{-4}	pass
2	endpoint deviatoric-strain relative L^2	4.31×10^{-3}	diagnostic
2	boundary profile relative L^2	5.23×10^{-4}	diagnostic
2	acceptance gate	–	pass

Table 12: Plasticity3D validation summary for same-case source-faithfulness (layer 1A) and the fixed-λ source-operator diagnostic (layer 2).

The resulting interpretation is limited but useful. Plasticity3D is validated here as a source-faithful maintained endpoint-surrogate workflow in Layer 1A, while Layer 2 supports the same endpoint surrogate under a fixed-load source-operator contract. Neither layer is a path-consistent incremental-plasticity history validation.

7 Performance and Solver Behavior

This section turns from problem definition to computational behavior. The benchmark families in Section 5 established the governing equations, discrete energies, and representative states, and Section 6 separated the validation and external-comparison evidence. Here we focus on parallel timings, scaling trends, solver behavior, and the comparative cost of alternative derivative or discretization choices. Evidence from older timing studies is retained when it is still informative for workflow viability, but it is labeled as timing evidence rather than merged into the validation story.

7.1 p-Laplace

The p-Laplace problem definition and discrete energy are given in Section 5.1 and equation (7). In this clean scalar setting, the numerical story is easiest to interpret: exact element Hessians remain effective in the reported distributed runs, while local-SFD reaches nearly identical discrete energies at higher wall time.

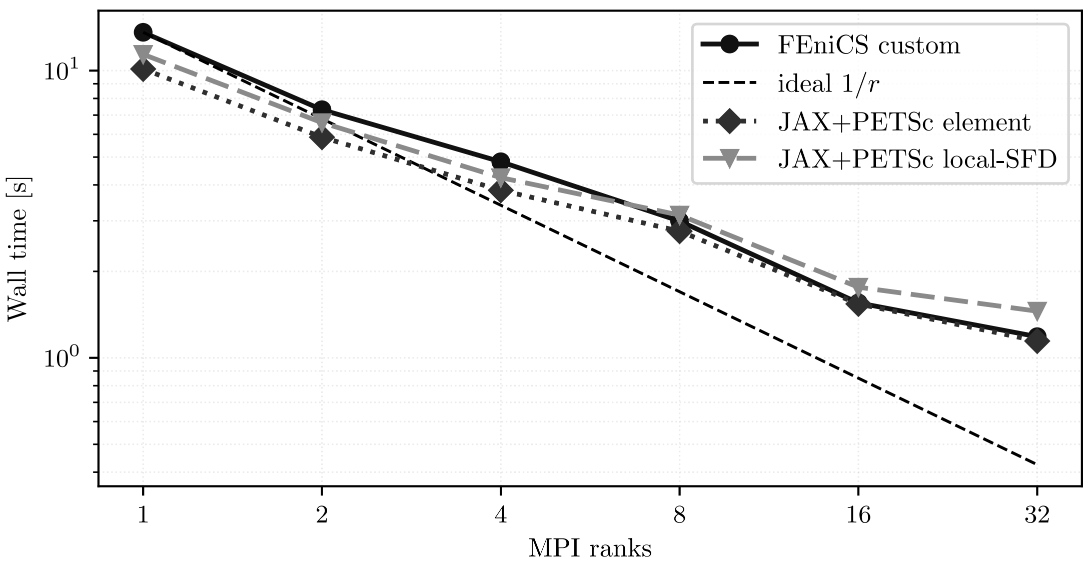


Figure 21: p-Laplace strong scaling on the finest maintained grid. All curves correspond to the same discrete problem from equation (7).

7.2 Ginzburg–Landau

The Ginzburg–Landau family from Section 5.2 and equation (11) shows the same high-level derivative story under a more indefinite linearization. The important computational message is not a dramatic separation between exact and approximate second-order routes, but that the maintained distributed solver remains convergent on this nonconvex scalar energy with the same sparse-infrastructure contract.

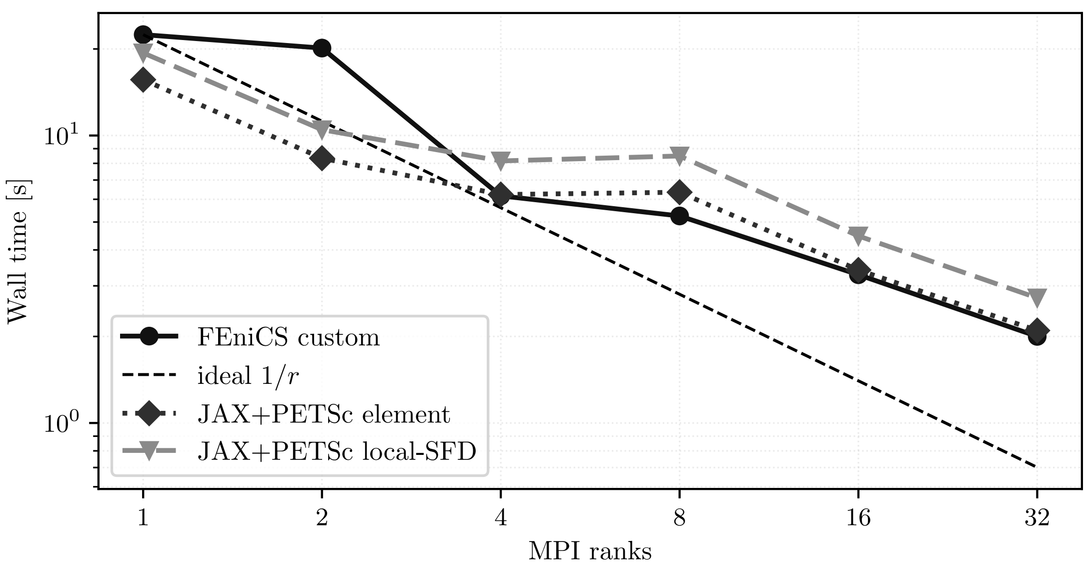


Figure 22: Ginzburg–Landau strong scaling for the maintained fine-grid case governed by equation (11).

7.3 Hyperelasticity

The hyperelasticity problem definition, boundary data, and terminal energies are given in Section 5.3 and equation (15). Computationally, this is the maintained benchmark in which globalization matters more than derivative-route choice: the distributed implementation completes the full load path and preserves the terminal energy trend from Section 5.3. The external JAX-FEM companion comparison is reported separately in Section 6.1; the figure below reports only distributed scaling for the repository load path.

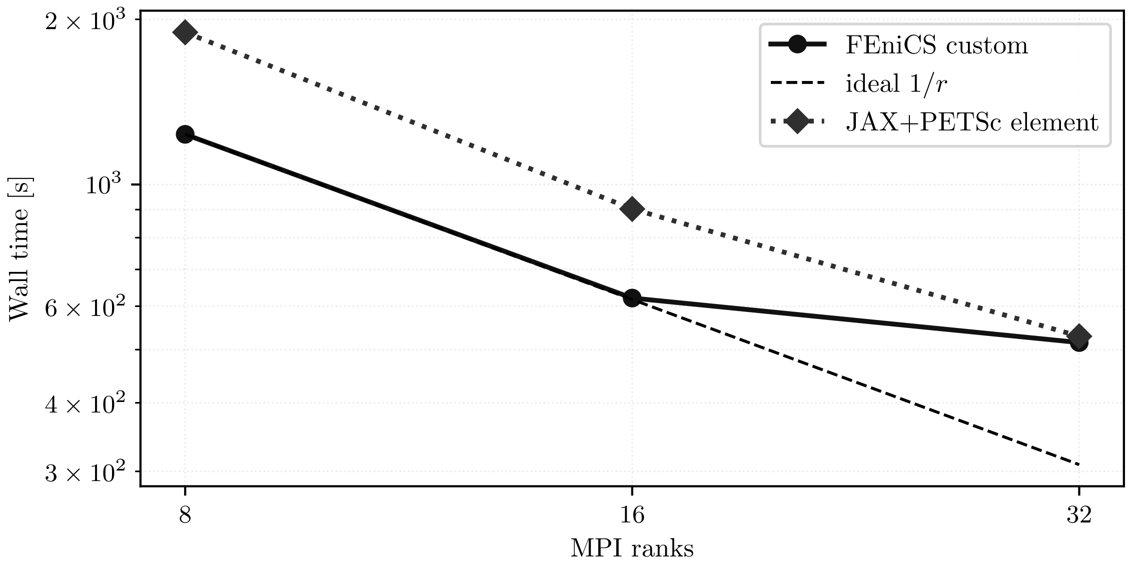


Figure 23: Hyperelasticity strong scaling for the 24-step load path defined by equation (15).

The larger fixed-work Karolina experiment isolates the first level-5 load step after the rank-local HyperElasticity implementation. The Karolina CPU nodes used here are IT4I standard CPU nodes with two AMD EPYC 7H12 64-core processors and 256 GB RAM [37]; on the tested nodes Slurm exposes the 128 cores as eight NUMA domains with 16 cores each. The runs use one MPI rank per core, contiguous CPU-ID binding, and local memory binding. The mesh is generated procedurally from the uniform beam hierarchy, and the element path assembles from rank-local elements with local PETSc COO preallocation and point-to-point overlap exchange.

The nonlinear solve is the same trust-region Newton path used for the benchmark in section 5.3. The linear subproblems use STCG with a same-hierarchy PMG preconditioner, Chebyshev/Jacobi smoothing, and coarsest level L_2. The coarse solve is PETSc redundant LU with MUMPS: the partial one-node rows use one redundant group over the active ranks, while the multi-node rows use one 128-rank redundant group per node. Figure 24 and table 13 report the application-reported solver total time, not Slurm elapsed time. All rows reach the same first-step energy to the precision shown in the raw outputs, with 21 Newton iterations and 43 Krylov iterations.

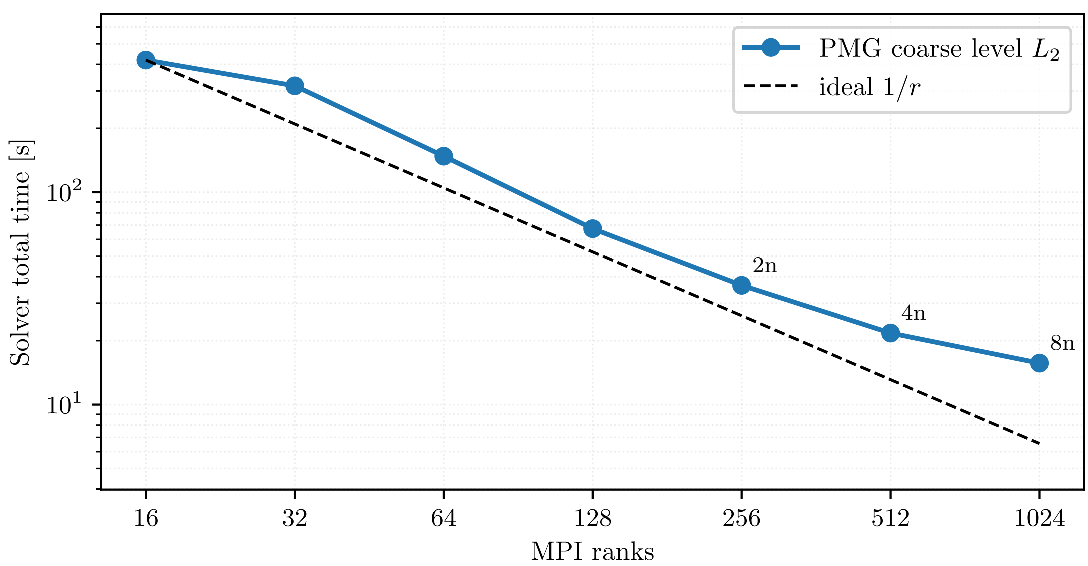


Figure 24: Karolina level-5 HyperElasticity first-step fixed-work scaling with the PMG coarse level fixed at L_2. Multi-node annotations show node counts.

Ranks	Nodes	Coarse groups	Ranks/group	Solver total [s]	First step [s]	Newton iters	Krylov iters
16	1	1	16	419	367	21	43
32	1	1	32	317	280	21	43
64	1	1	64	148	132	21	43
128	1	1	128	67.4	58.4	21	43
256	2	2	128	36.4	30.7	21	43
512	4	4	128	21.7	17.7	21	43
1024	8	8	128	15.7	12.3	21	43

Table 13: Karolina level-5 HyperElasticity first-step timings used in figure 24.

7.4 Plasticity2D

The Plasticity2D model and Davis-B reduction are defined in Section 5.4 and equations (18) and (19). For this family, the most informative computational evidence is not a single strong-scaling curve but the solver-engineering material collected in Section 8: same-mesh PMG choices, fixed-work runs, and source continuation behavior. We therefore keep the within-benchmark resolution trend in Figure 13 and defer the supporting solver data to the later evidence section.

7.5 Plasticity3D

The governing problem, Davis-B reduction, scalar constitutive potential, and deviatoric-strain norm are defined in Section 5.5 and equations (22), (24), (29) and (30). The validation limits for this endpoint surrogate are reported in Section 6.2. With that interpretation fixed, this section asks two performance questions: which second-order route is preferred on the locked flagship case, and how the maintained surrogate behaves across degree, resolution, and rank count.

Derivative-route ablation. The first question is which second-order route has the most useful cost-quality profile for the promoted maintained workflow. On the locked $P_4(L_1)$ case at $\lambda_{sr} = 1.5$ and 8 MPI ranks, all three routes converge to essentially the same terminal state, with identical Newton and total Krylov counts. The differences are therefore cost differences, not state-quality differences: constitutive AD has the lowest median wall time on this locked case, with median wall time 222 s and median solve time 191 s, compared with 288 s/255 s for element AD and 372 s/253 s for colored SFD.

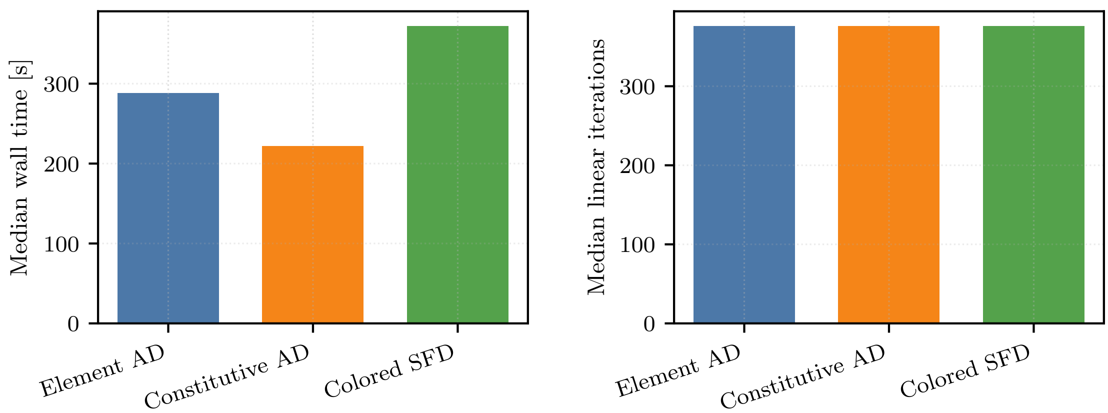
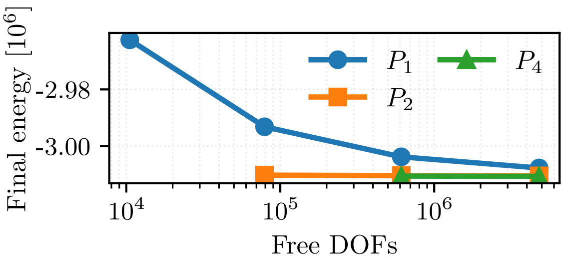


Figure 25: Locked Plasticity3D derivative-route ablation on $P_4(L_1)$, $\lambda_{sr} = 1.5$, and 8 MPI ranks.

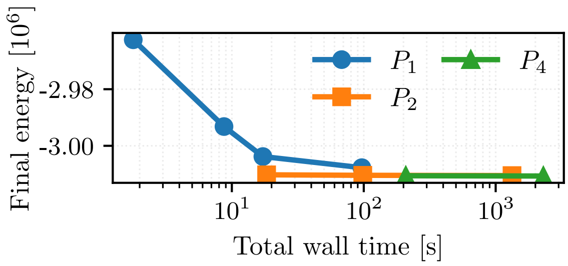
Route	Wall time [s]	Solve time [s]	Newton iters	Krylov iters	Energy	ω	u_{max}
Element AD	288	255	8	376	−3,010,861	6,027,722	0.731,848
Constitutive AD	222	191	8	376	−3,010,861	6,027,722	0.731,848
Colored SFD	372	253	8	376	−3,010,861	6,027,722	0.731,848

Table 14: Plasticity3D derivative-route ablation on the locked $P_4(L_1)$, $\lambda_{sr} = 1.5$, 8-rank case.

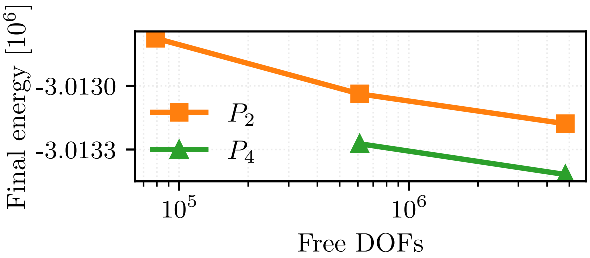
Maintained discretization and scaling. The remaining figures in this subsection answer a different question: how the maintained surrogate behaves across degree/resolution choices, and how the recommended maintained stack scales on the separate promoted $P_4(L_2)$ timing studies.



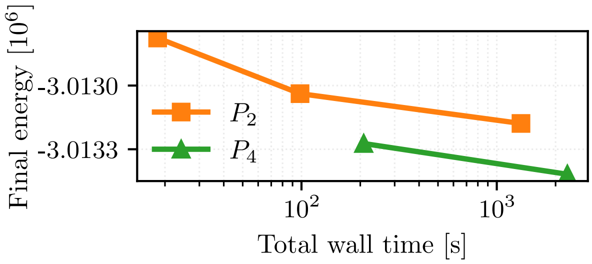
(a) All degrees: final energy versus free DOFs.



(b) All degrees: final energy versus total wall time.



(c) $P2/P4$ zoom: final energy versus free DOFs.



(d) $P2/P4$ zoom: final energy versus total wall time.

Figure 26: Corrected glued-bottom maintained Plasticity3D degree-versus-resolution study at $\lambda_{\mathrm{sr}} = 1.55$. The top row shows the full $P_1/P_2/P_4$ comparison, while the bottom row zooms to P_2/P_4 so the near-overlap hidden by the coarse P_1 points becomes visible. The left column plots final values of equation (22) against free DOFs; the right column plots the same converged energies against end-to-end wall time. All timings are fair 32-rank maintained runs.

The zoom row in Figure 26 is essential because the full figure is dominated by coarse P_1 variation. Under the corrected glued-bottom constraint, the first matched comparison is $P_4(L_1)$ versus $P_1(L_3)$ at 610,964 free DOFs. The second matched comparison is $P_1(L_4)$, $P_2(L_3)$, and $P_4(L_2)$ at 4,801,816 free DOFs. The scientific message is therefore that the maintained solver reaches similar terminal energies along both p- and h-refined paths, while higher order on coarser meshes buys lower energy at matched DOFs only by paying a clear wall-time premium.

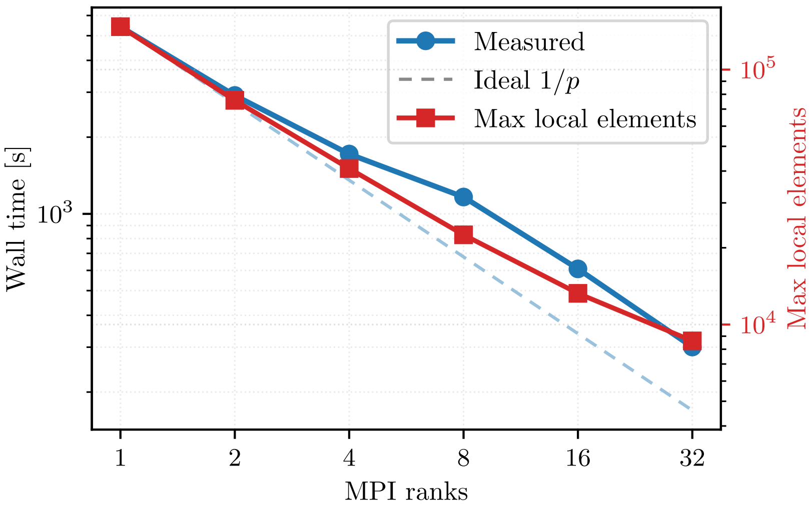


Figure 27: Historical strong scaling for the promoted Plasticity3D maintained stack: constitutive-AD local assembly with same-mesh PMG on $P_4(L_2)$ at $\lambda_{\mathrm{sr}} = 1.0$. This is a separate timing study from the glued-bottom $\lambda_{\mathrm{sr}} = 1.55$ results shown in figures 14, 15 and 26.

Ranks	Wall time [s]	Solve time [s]	Speedup	Efficiency	Newton iters	Krylov iters
1	5424	5176	1	1	17	114
2	2914	2777	1.861	0.931	17	114
4	1717	1634	3.16	0.79	17	114
8	1164	1107	4.658	0.582	17	114
16	608	575	8.916	0.557	17	114
32	300	284	18.053	0.564	17	114

Table 15: Historical Plasticity3D strong scaling for constitutive-AD local assembly with same-mesh PMG on $P_4(L_2)$ at $\lambda_{\mathrm{sr}} = 1.0$. These rows are useful as solver evidence, but they should not be read as a direct timing continuation of the glued-bottom $\lambda_{\mathrm{sr}} = 1.55$ benchmark figures.

Figure 27 and table 15 give the paper's largest maintained distributed mechanics timing result. This scaling evidence uses the promoted $P_4(L_2)$ mesh family but a different load factor, $\lambda_{\mathrm{sr}} = 1.0$, so it complements rather than extends the glued-bottom $\lambda_{\mathrm{sr}} = 1.55$ benchmark figures above. Within that timing study, the reported nonlinear and linear iteration counts remain matched on 1/2/4/8/16/32 ranks, while the end-to-end time drops from 5424 s on one rank to 300 s on 32 ranks. Additional source-vs-local and continuation evidence is reported later in Section 8.

The converged glued-bottom $P_4(L_2)$, $\lambda_{\mathrm{sr}} = 1.55$ scaling run repeats the benchmark with the current constitutive-AD local assembly and MUMPS-backed PMG stack. The PMG hierarchy is the same-mesh $P_4 \to P_2 \to P_1$ hierarchy, the nonlinear method is Armijo Newton, and the Krylov tolerance is 1×10^{-2}. The nonlinear stop requires both relative correction below 0.002 and gradient norm below one percent of the first Newton gradient. Figure 28 and table 16 overlay the local workstation sweep with the Karolina CPU-node sweep at 16 MPI ranks per node. All plotted rows converge in 29 Newton iterations and 1240 total Krylov iterations to the same energy and gradient norm to the precision shown in the table, so the timing comparison is not confounded by different nonlinear work.

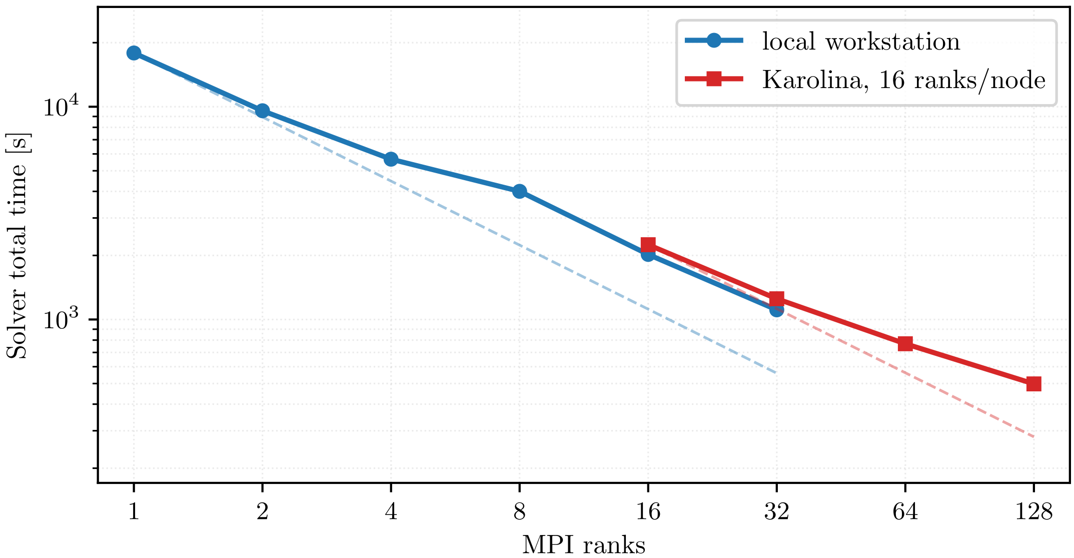


Figure 28: Local and Karolina converged scaling for the glued-bottom Plasticity3D $P_4(L_2)$ case at $\lambda_{\mathrm{sr}} = 1.55$. Dashed references are ideal $1/r$ lines anchored separately to the first point of each platform.

Platform	Ranks	Nodes	Solver total [s]	Solve [s]	Krylov iters	Energy	$\|g\|$
local workstation	1	–	17,890	17,615	1240	−3,013,348	10.594
local workstation	2	–	9574	9412	1240	−3,013,348	10.594
local workstation	4	–	5667	5564	1240	−3,013,348	10.594
local workstation	8	–	4003	3947	1240	−3,013,348	10.594
local workstation	16	–	2022	1993	1240	−3,013,348	10.594
local workstation	32	–	1110	1093	1240	−3,013,348	10.594
Karolina	16	1	2246	2204	1240	−3,013,348	10.594
Karolina	32	2	1251	1227	1240	−3,013,348	10.594
Karolina	64	4	769	754	1240	−3,013,348	10.594
Karolina	128	8	498	488	1240	−3,013,348	10.594

Table 16: Converged timings for the Plasticity3D $P_4(L_2)$, $\lambda_{\mathrm{sr}} = 1.55$ run in figure 28. Local rows are single-workstation MPI runs; Karolina rows use 16 ranks per CPU node.

The distributed layout diagnostics in Table 17 explain why the observed scaling cannot follow the ideal reference indefinitely. The maximum local overlap vector shrinks with rank count, but the sum of rank-local overlap DOFs increases from about the owned problem size on one rank to more than four times the owned size on 128 ranks. This overlap replication comes from subdomain boundaries and high-order element support, so it is a decomposition cost rather than duplicated global mesh storage.

Platform	Ranks	Nodes	Max local DOFs	Overlap / owned DOFs
local workstation	1	–	4,859,595	1.01
local workstation	2	–	2,495,265	1.05
local workstation	4	–	1,284,351	1.11
local workstation	8	–	669,783	1.21
local workstation	16	–	351,219	1.41
local workstation	32	–	182,613	1.83
Karolina	16	1	351,219	1.41
Karolina	32	2	182,613	1.83
Karolina	64	4	91,821	2.66
Karolina	128	8	50,637	4.31

Table 17: Distributed layout diagnostics for the converged Plasticity3D scaling run. "Max local DOFs" is the largest local-overlap vector size on any rank; "Overlap / owned DOFs" is the summed rank-local overlap DOF count divided by the global owned free-DOF count.

7.6 Topology Optimization

The reduced design problem, mechanics subproblem, and objective history are defined in Section 5.6 and equations (32) and (33). The numerical question here is distributed outer-loop stability: the mechanics solve and the design update must remain synchronized under continuation in p and move-regularized updates. Because the outer-loop work can vary with the continuation schedule, the timing evidence here should be read as end-to-end workflow scaling rather than fixed-work strong scaling.

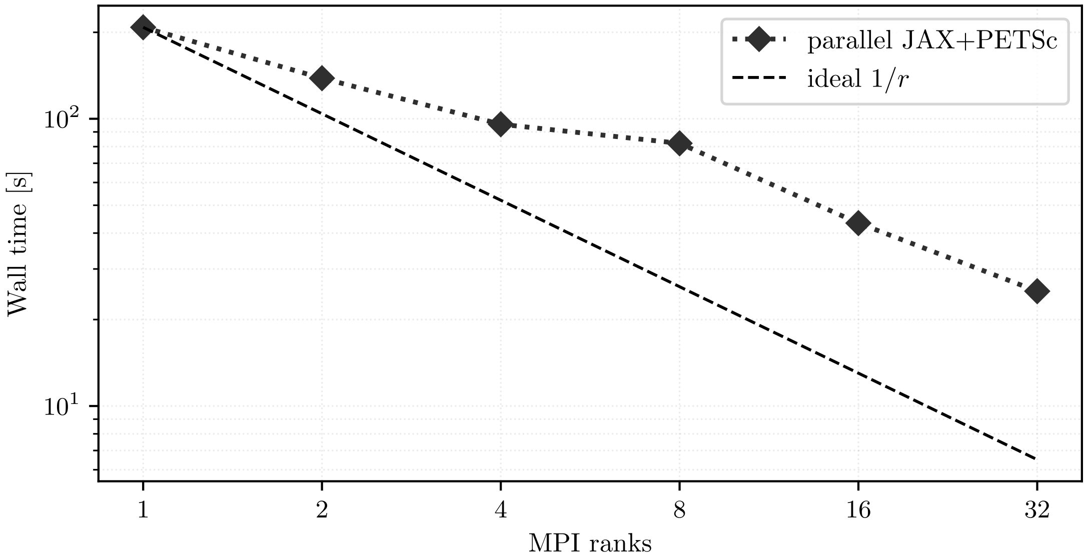


Figure 29: Topology end-to-end parallel timing trend for the maintained fine-grid run. The plotted wall times correspond to the coupled reduced-design workflow defined by equation (33).

Ranks	Wall time [s]	Solve time [s]	Outer iters	p	Compliance	Volume fraction
1	208	200	65	4.2	9.1559	0.3884
2	138	134	72	5.6	8.9473	0.3932
4	95.7	92.8	69	4.6	9.1684	0.385
8	82.1	80.1	67	7	9.2178	0.3932
16	43.3	41.9	66	5.8	9.6859	0.3798
32	25.1	23.7	66	6.6	9.9074	0.3749

Table 18: Parallel topology summary for the maintained fine-grid study, reported for the reduced objective in equation (33).

The topology benchmark completes the paper's computational picture. Here the mainline JAX+PETSc path is less about exact Hessians than about stable distributed design-mechanics coupling, while pure JAX remains the compact serial reference path discussed earlier in Table 4.

8 Additional Solver Evidence

This section collects supporting solver evidence that is methodologically useful but too implementation-specific for the main benchmark narratives.

8.1 Plasticity3D Source-Assembly and Fixed-Operator PMG Comparisons

The primary Plasticity3D numerical story in Section 7.5 is the maintained local path. The tables below add two engineering comparisons on the same discrete problem from equation (22): a matched source-assembly comparator and a fixed source-operator PMG variant.

Ranks	Constitutive wall [s]	Source wall [s]	Constitutive solve [s]	Source solve [s]	Ratio
4	1717	951	1634	890	1.805
8	1164	697	1107	656	1.671
16	608	374	575	350	1.628
32	300	277	284	260	1.085

Table 19: Matched local-vs-source comparison for the same local PMG solver stack. At each listed rank the local and source rows converge in the same number of nonlinear and linear iterations, so this is a timing-only comparison on equation (22).

Ranks	Route	Wall time [s]	Newton iters	Krylov iters	Final gradient norm	Status
4	constitutive-AD PMG solver	2116	24	204	6.53×10^{-3}	completed
4	source-assembly PMG variant	988	21	210	2.49×10^{-3}	completed
8	constitutive-AD PMG solver	1874	33	238	2.53×10^{-3}	completed
8	source-assembly PMG variant	898	27	246	9.53×10^{-3}	completed
16	constitutive-AD PMG solver	1304	50	196	2.21×10^{2}	failed
16	source-assembly PMG variant	601	50	177	7.35×10^{2}	failed
32	constitutive-AD PMG solver	348	29	250	2.24×10^{-3}	completed
32	source-assembly PMG variant	398	50	193	8.25×10^{1}	failed

Table 20: Fixed source-operator PMG alternative on the same Plasticity3D problem. The status column distinguishes completed rows from rank counts that hit the Newton cap before reaching the promoted gradient target.

These tables support two narrower statements. First, source assembly has lower reported wall time than maintained local assembly when both are embedded in the same PMG solver stack, although the gap narrows at higher rank counts. Second, the fixed source-operator PMG profile is not the preferred maintained default in the present study: it may appear favorable on some rows, but it does more nonlinear and Krylov work and converges less consistently across the reported runs.

8.2 Source continuation evidence

The legacy source-continuation path is not the mainline method of the paper, but it is still useful because it isolates how much preconditioner policy matters even when the outer continuation logic is held fixed.

Policy	Ranks	Runtime [s]	Init Krylov iters	Continuation Krylov iters	Final λ
fixed PMG-shell smoother	8	489	58	733	1.568,403
fixed PMG-shell smoother	32	198	60	829	1.568,334

Table 21: Source continuation with the fixed PMG-shell smoother policy at 8 and 32 ranks. This is supporting solver evidence only; it is not part of the paper's main maintained production story.

8.3 Code, data, and reproducibility availability

The implementation discussed in this paper is available at `https://github.com/Beremi/fenics_nonlinear_energies`. The manuscript is generated from the repository-local paper pipeline. The table, figure, literature-source, asset-validation, and reproducibility-note scripts live in `paper/scripts`. The generated manuscript assets live in `paper/tables/generated` and `paper/figures/generated`. The raw benchmark summaries used by those scripts are kept under `artifacts/raw_results`. The paper build validates that the generated assets are present before compiling the PDF. A citable source or artifact archive DOI should be supplied in the final submission metadata if required by the target journal; this journal-neutral draft does not assert one.

9 Discussion

The results support four main conclusions.

Constitutive AD is the preferred derivative route on the flagship Plasticity3D case. Figure 25 and table 14 show a clean cost-quality picture on the locked $P_4(L_1)$, $\lambda_{\mathrm{sr}} = 1.5$, 8-rank case. All three routes reach essentially the same terminal state with the same nonlinear and Krylov counts, so the comparison is about cost rather than about different converged solutions. On that evidence, constitutive AD is the preferred maintained route: it preserves the scalar-energy implementation contract while reducing wall time substantially relative to full element AD and colored SFD.

Validation has to be separated by claim level. The Plasticity3D validation ladder in Figure 20 and table 12 changes how the paper should phrase its mechanics claims. Same-case source-faithfulness to the source-style surrogate implementation is strong, and that is an important result for software credibility. The fixed-lambda source-operator comparison also satisfies the configured endpoint field-agreement thresholds once the source boundary condition is matched to the maintained glued-bottom mask. The right interpretation is therefore stronger but still scoped: the maintained implementation faithfully reproduces the intended endpoint-surrogate workflow, yet the present evidence does not justify describing that surrogate as a path-consistent incremental elastoplastic history solve or as a verified numeric reproduction of the Sysala source-family papers.

Matched external baselines are useful when the contract is narrow. The hyperelastic companion comparison against JAX-FEM in Figure 19 and table 11 shows that a narrow, fairness-first external baseline can survive review-level scrutiny. Here the mesh and prescribed displacement schedule are matched, and the terminal states are post-compared with the repository energy; the resulting energy and field differences are very small. The comparison is useful precisely because it is narrow and hyperelastic-only. The paper should not generalize it into a broad ranking between frameworks or into plasticity validation, but it does strengthen the manuscript by showing that at least one recent external implementation can be matched on a shared hyperelastic problem contract.

Fairness and limitations remain part of the scientific result. Several limitations should stay explicit. The historical Plasticity3D strong scaling study at $\lambda_{sr} = 1.0$ remains workflow evidence, while the newer glued-bottom $\lambda_{sr} = 1.55$ scaling sweep is the converged comparison tied to the maintained benchmark. The topology study is reported as end-to-end workflow scaling rather than fixed-work strong scaling. Some source/local PMG studies remain solver-engineering diagnostics rather than fully symmetric scientific baselines. None of these caveats erase the paper's main contribution, but they do define its proper scope: a maintained software-and-methods workflow with explicit derivative-route choices, explicit validation surfaces, and explicit fairness boundaries.

The broader methodological lesson is that differentiable FEM becomes most scientifically useful when local differentiation and sparse solver engineering are designed together. Automatic differentiation alone does not settle the behavior of large nonlinear FEM workflows; globalization, sparse ownership, preconditioning, and the interpretation of surrogate models matter just as much. Within that scoped claim, `fenics_nonlinear_energies` contributes a reproducible maintained workflow rather than a universal differentiable-PDE platform or a new constitutive theory.

10 Conclusion

This paper presented `fenics_nonlinear_energies` as a software-and-methods workflow for nonlinear finite-element solves and optimization with a maintained JAX+PETSc production path. The central contribution is not a new constitutive theory or a universal differentiable-PDE abstraction, but a maintained implementation pattern in which local JAX differentiation, sparse PETSc nonlinear solves, and multiple second-order routes are kept together across a heterogeneous benchmark suite.

The Plasticity3D evidence is intentionally narrower than a broad mechanics claim. Same-case source-faithfulness to the source-style surrogate implementation is strong, and the fixed-lambda source-operator comparison passes the configured gated quantities after matching the glued-bottom boundary contract, with additional field-agreement diagnostics reported separately. These results support the maintained endpoint-surrogate workflow for this benchmark family, while still not supporting constitutive-equivalence language or path-consistent incremental-history claims. Constitutive AD emerges as the preferred cost-quality route on the locked flagship case. On Hyperelasticity, the companion comparison against JAX-FEM shows that the maintained code can agree very closely with a recent external implementation when the mesh and prescribed displacement schedule are matched and terminal states are post-compared with the repository energy. The promoted Plasticity3D timing study and the distributed topology workflow then show that the same codebase remains operational beyond small validation comparisons.

Future work should follow the boundaries made visible by these results:

- extend the validation surface for path-dependent mechanics beyond the present surrogate-versus-source-operator comparison, including a true incremental-history reference;
- add more fairness-first external baselines on tightly matched problem contracts;
- continue reducing linear-setup and multigrid costs in the flagship 3D mechanics workflows.

The next scientific step is not to overclaim this implementation, but to turn this maintained workflow into a stronger reusable template for additional nonlinear multiphysics and design problems with equally explicit validation and fairness criteria.

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